

CS 736

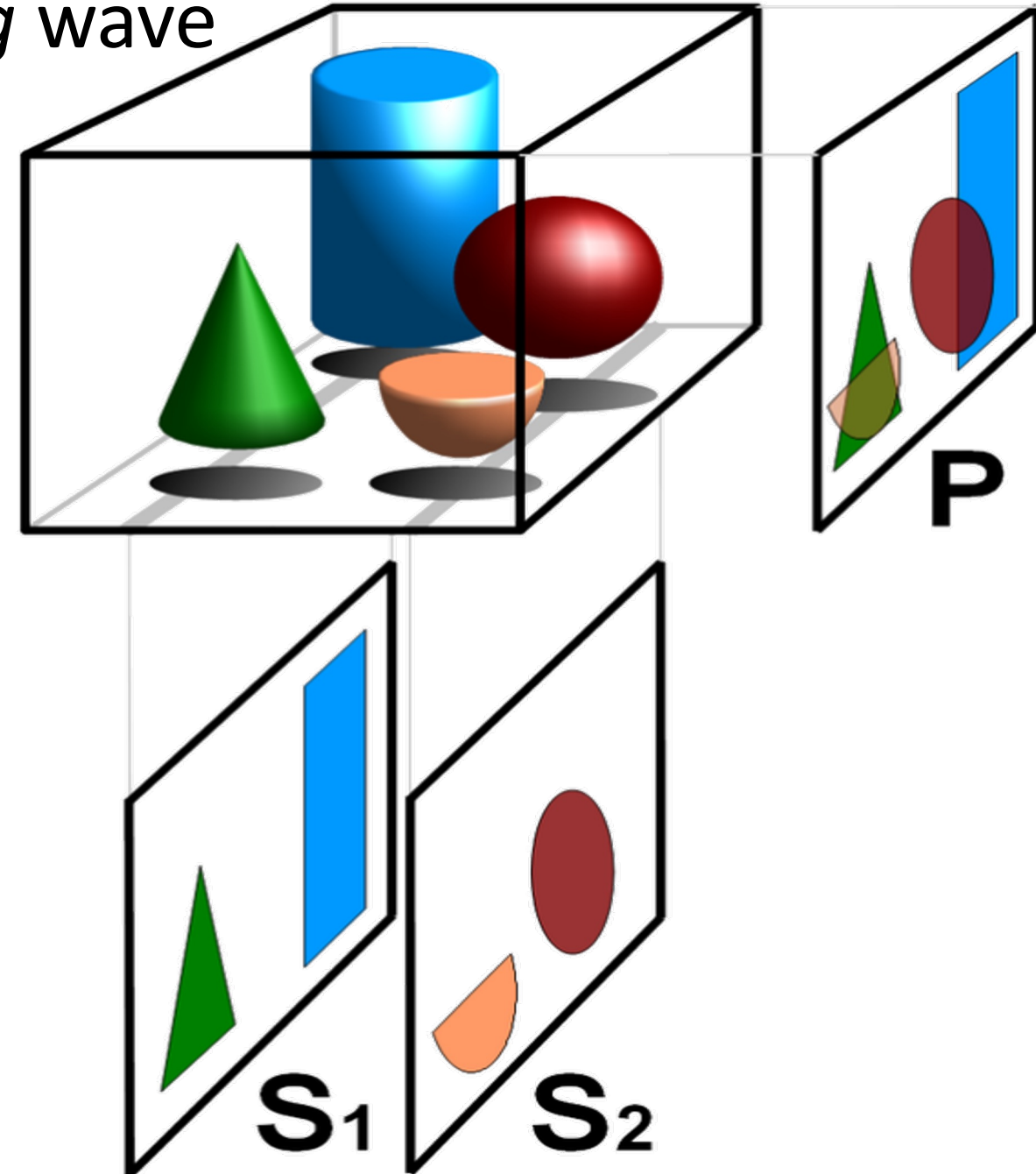
Medical Image Computing (MIC)

Magnetic Resonance Imaging (MRI)

Suyash P. Awate

Imaging

- Tomography: imaging by sections or sectioning/slicing, through the use of any kind of *penetrating* wave
 - P = projection
 - S = section/slice
 - Greek word
 - *tomē* ("cut"), *tomos* ("part" or "section")
 - *graphein* ("to write")
 - Display a section
- Tomograph = device used for tomography
- Tomogram = image produced

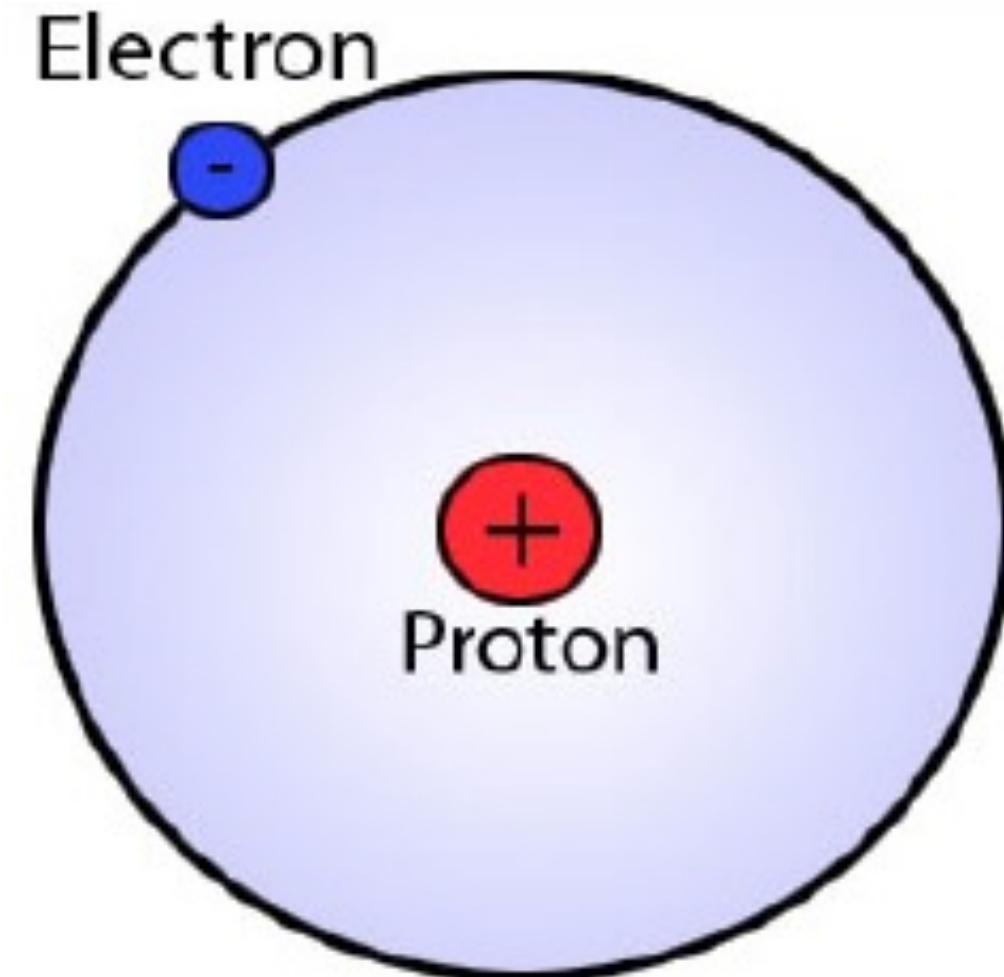


Imaging: Tomography

- Magnetic resonance tomography (MRT) / magnetic resonance imaging (MRI) → radio-frequency waves
- X-ray computed tomography (CT) → X rays
- Positron-emission tomography (PET) → gamma rays
- Optical tomography → infra-red light
- Ultrasound imaging → sound
- Microscopy (“3D optical projection”) → visible light
 - en.wikipedia.org/wiki/Digital_microscope#3D_measurement
 - depts.washington.edu/hplab/research/3d-optical-projection-tomographic-microscopy/

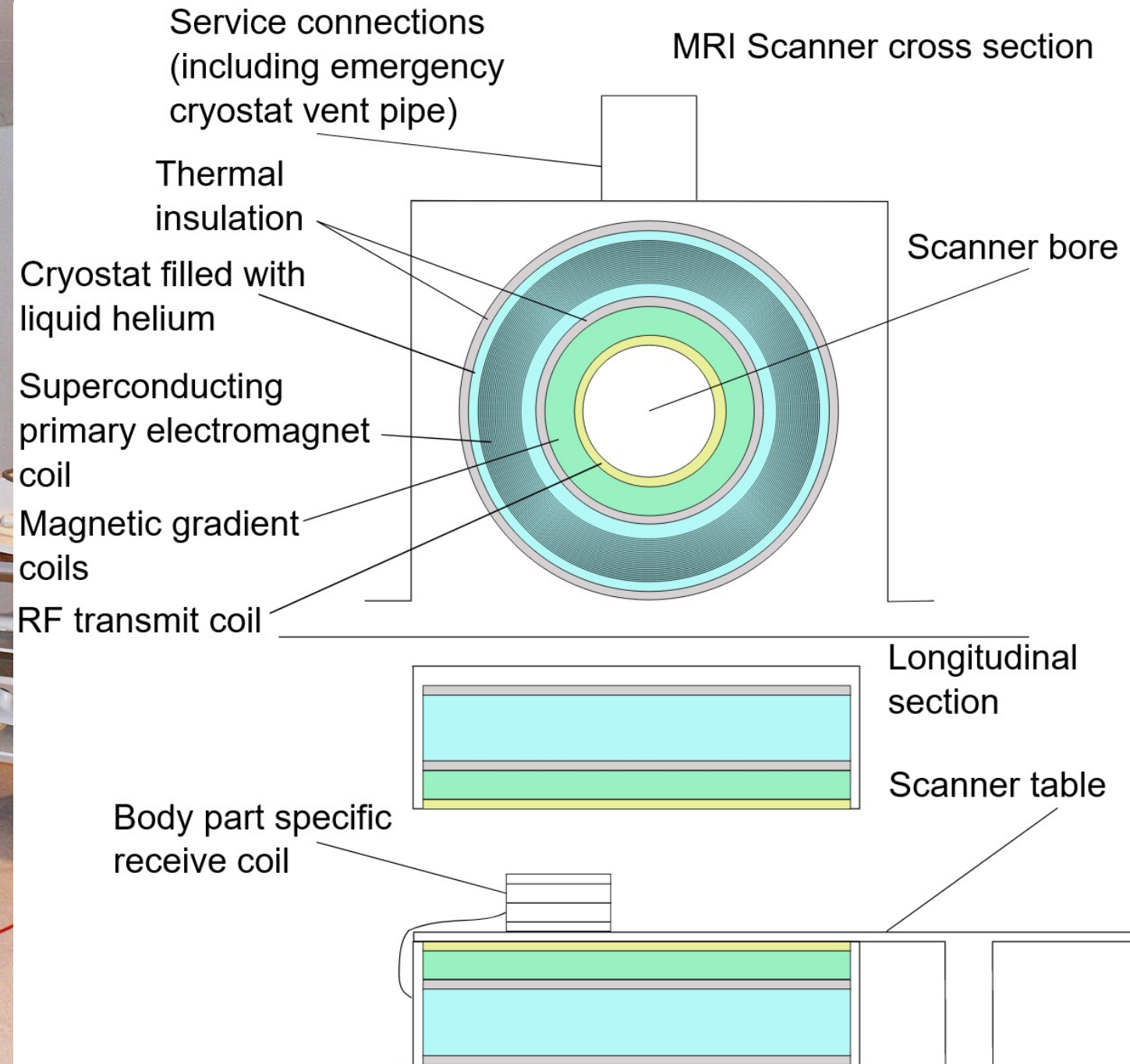
MRI

- The human body is, basically, water
 - Around 53% water by mass
 - Water content in:
Blood: 93%, Muscle: 70%, Heart: 80%, Liver: 70%, Kidneys: 80%,
Brain (Gray matter: 71%, White matter: 84%),
Bone: 12%
- The human body has lots of hydrogen
 - 11% hydrogen by mass
 - 65% Oxygen, 18% Carbon
 - 67% hydrogen by atomic percent
 - More hydrogen atoms than any other element



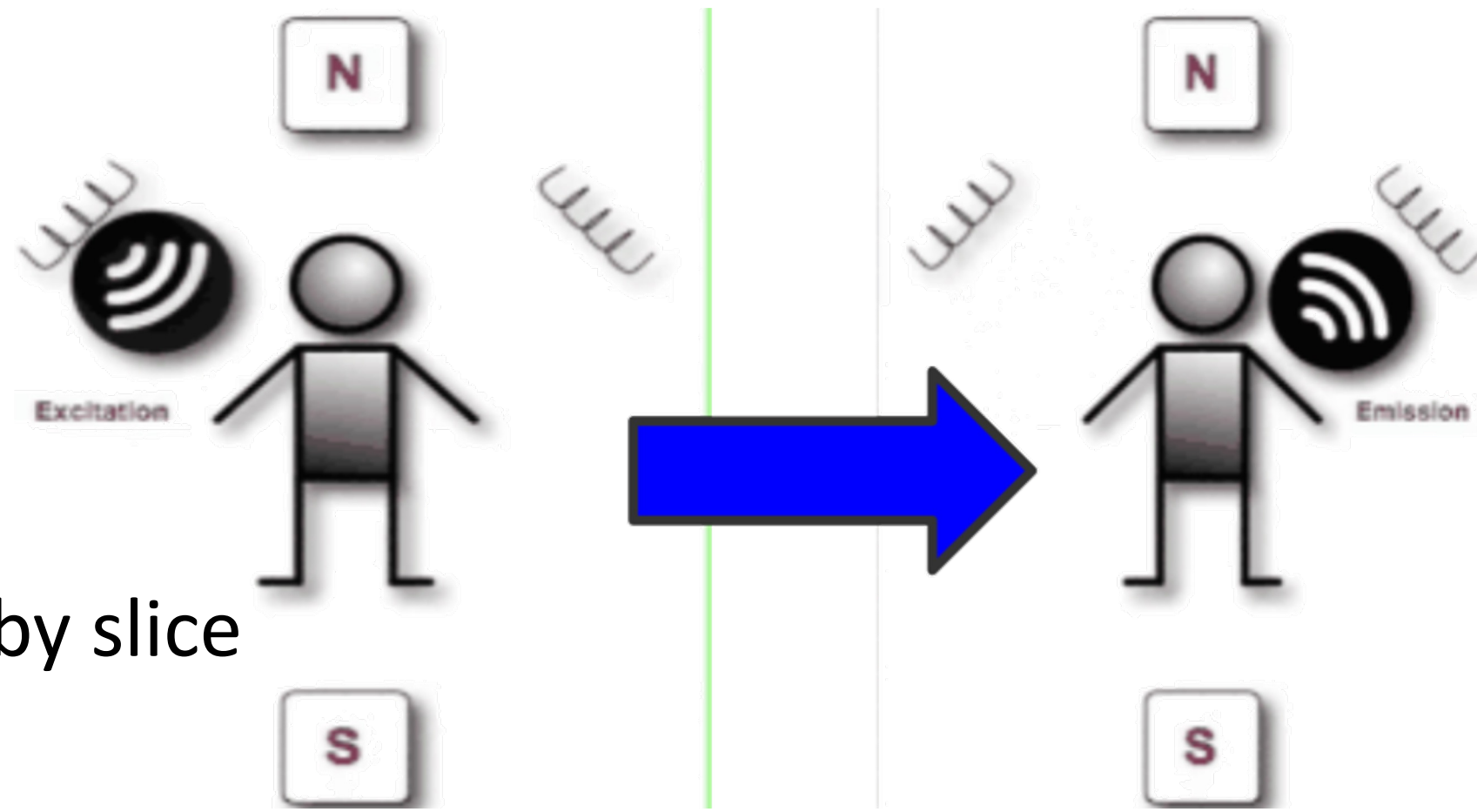
MRI: Machine

- en.wikipedia.org/wiki/Magnetic_resonance_imaging



MRI: Acquisition Process

- 1) Place patient inside a large strong magnetic field
 - How strong ? (unit for magnetic-flux density: tesla)
 - Earth's magnetic field on surface is 25-65 micro-tesla
 - Refrigerator magnet (iron alloy) → 5 milli-tesla
 - Magnetic field with clinical MRI → 1.5 tesla, 3 tesla, 7 tesla
- 2) Use additional time-varying & space-varying electromagnetic fields
 - Need precise control over fields, timing
- 3) Typically acquire data slice by slice



MRI: Acquisition Pr

- www.youtube.com/watch?v=4uzJPpC4Wuk



MRI: Acquisition Process

- www.youtube.com/watch?v=vn5H-6BbesQ



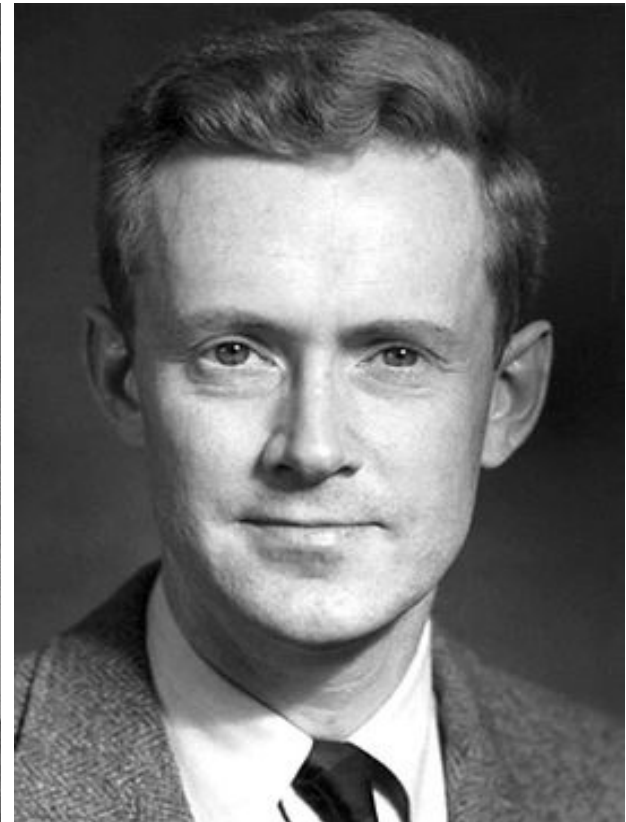
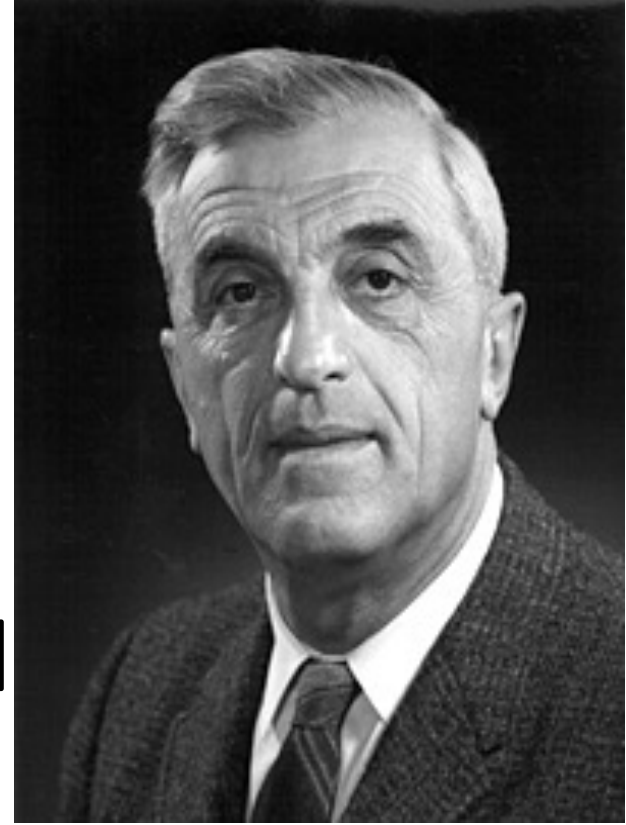
MRI: Acquisition Process



medicaldialogues.in/ge-pays-rs-1-crore-settlement-to-technician-who-got-stuck-in-mri-at-tata-memorial

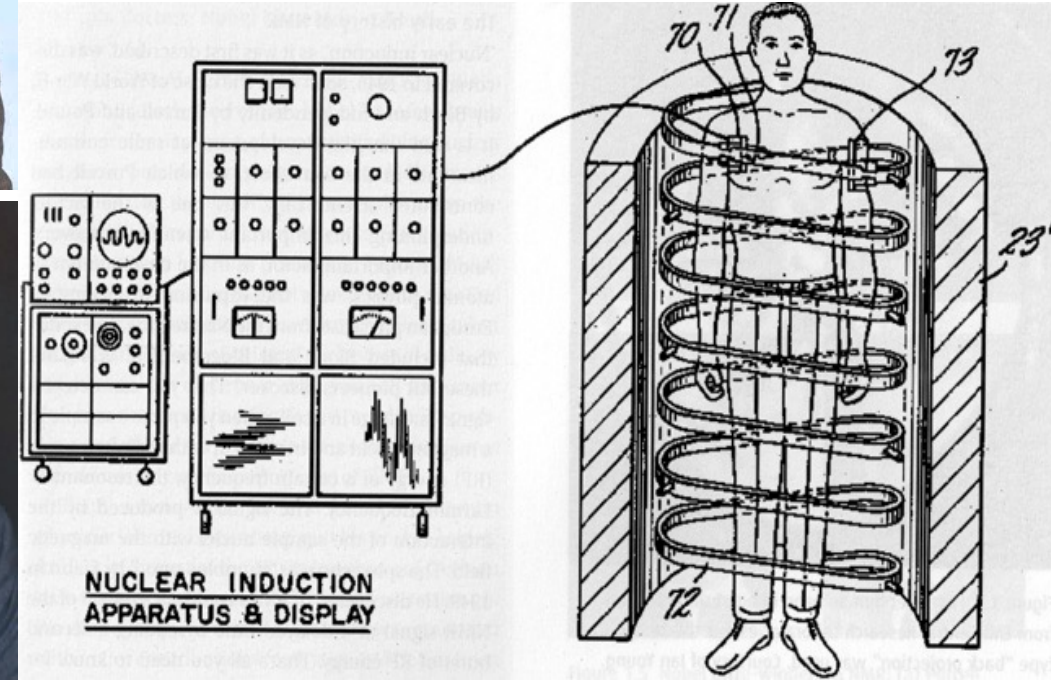
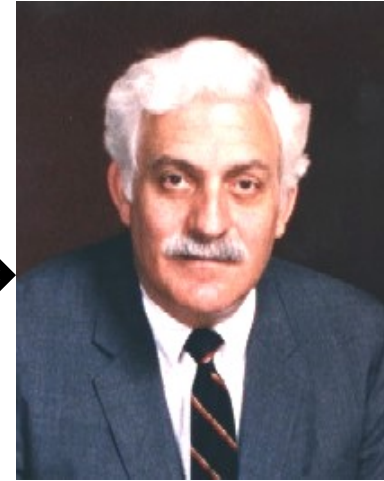
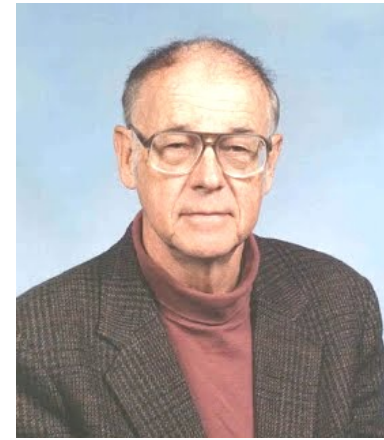
MRI: History

- Theory of nuclear magnetic resonance (NMR) 1946
 - Felix Bloch: classical physics
 - Edward Purcell: quantum physics
 - Interestingly:
Bloch was a trained quantum physicist [Advisor: Heisenberg]
Purcell was a trained classical physicist
- Bloch and Purcell shared 1952 Nobel prize in physics
- Werner Heisenberg
 - Nobel Prize in Physics for 1932 "for the creation of quantum mechanics"



MRI: History

- First 1D MR **signal** (1952)
 - Herman Carr's PhD work at Harvard
- 1st clinical application of NMR: 1971
 - Dr. Raymond Damadian (*Science*)
 - **Conceptualized** human-sized **scanner** in 1977, to spatially-localize disease →
- First **2D-slice** image (1970s)
 - Paul **Lauterbur** (Chemist), building on Herman Carr's work, Stony Brook University
 - Measured 2D projections of the NMR signal in a 3D object
- First mathematical analysis for **faster** imaging (1970s)
 - Sir Peter **Mansfield** (Physicist), University of Nottingham
 - **Slice selection**: frequency + phase **encoding** by spatial magnetic-field gradients



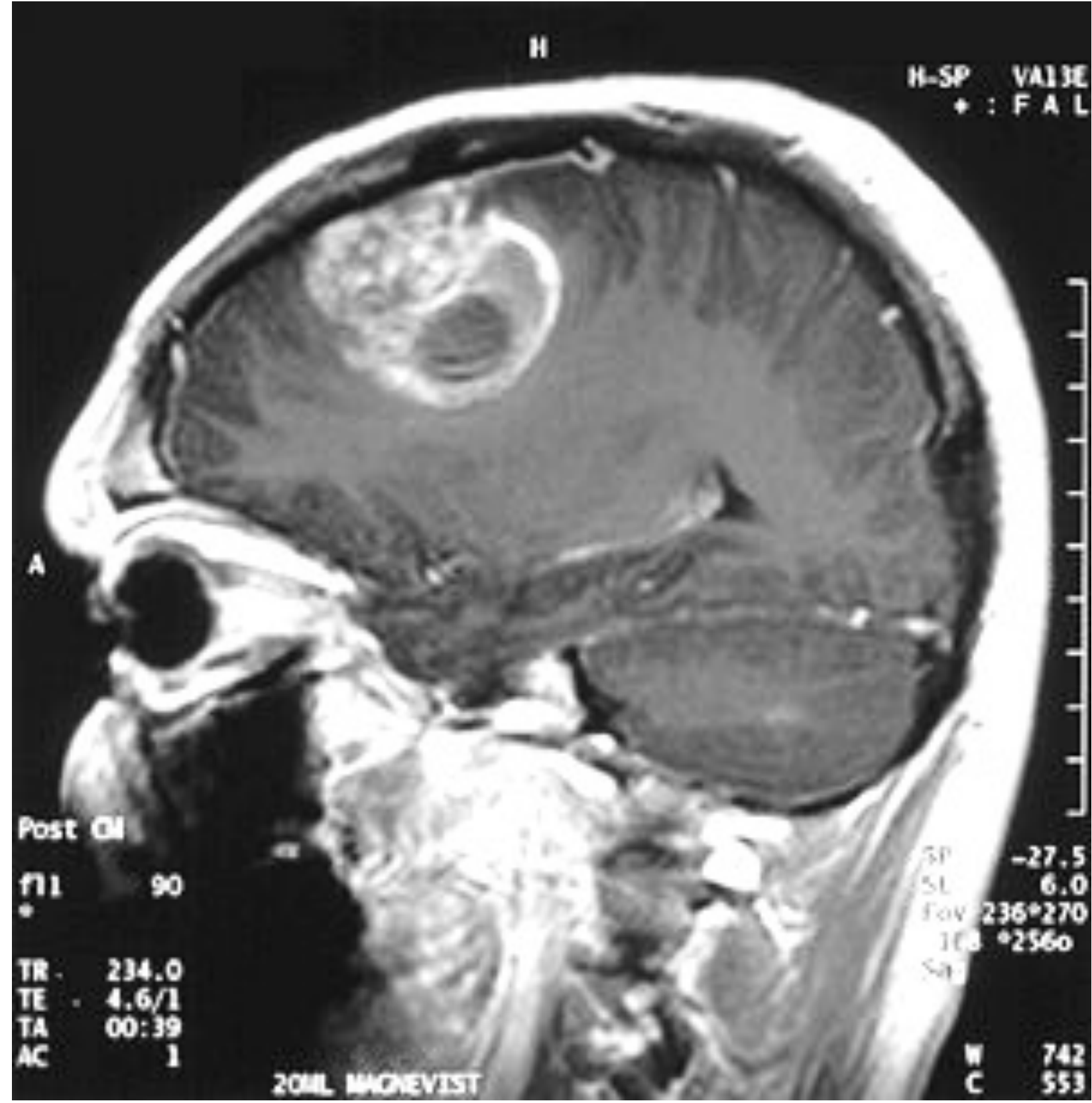
MRI: History

- Lauterbur & Mansfield got 2003 Nobel prize in Medicine & Physiology



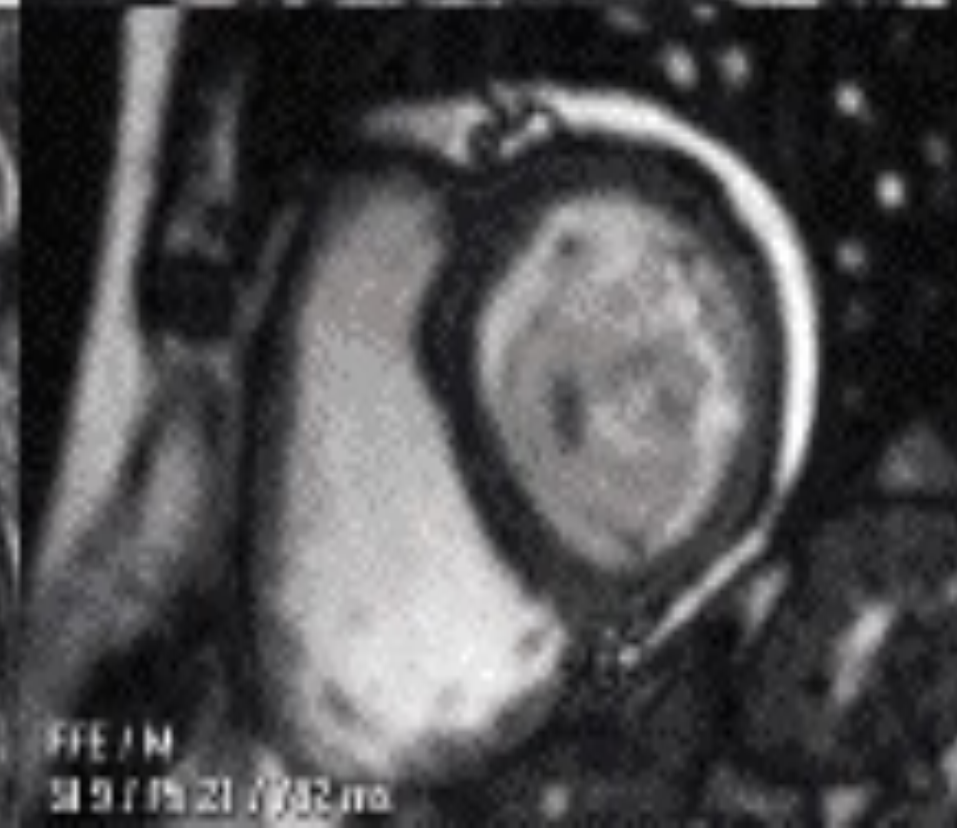
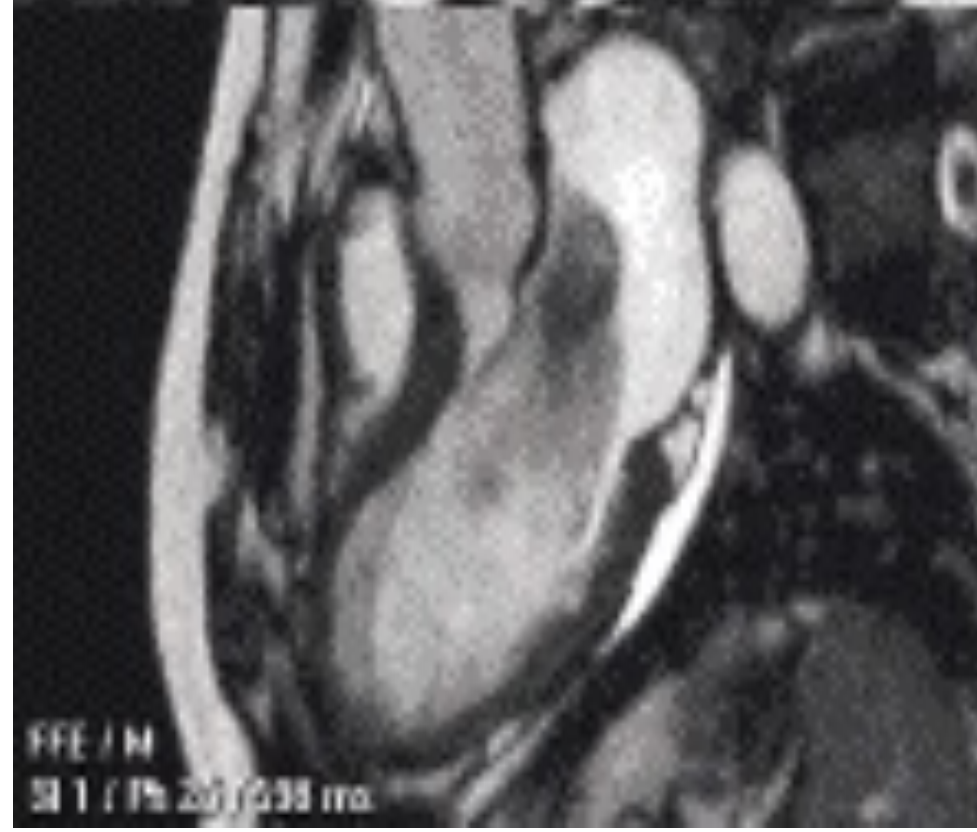
MRI

- Examples:
Brain MRI
 - Investigate dementia, cerebrovascular disease, epilepsy, tumors



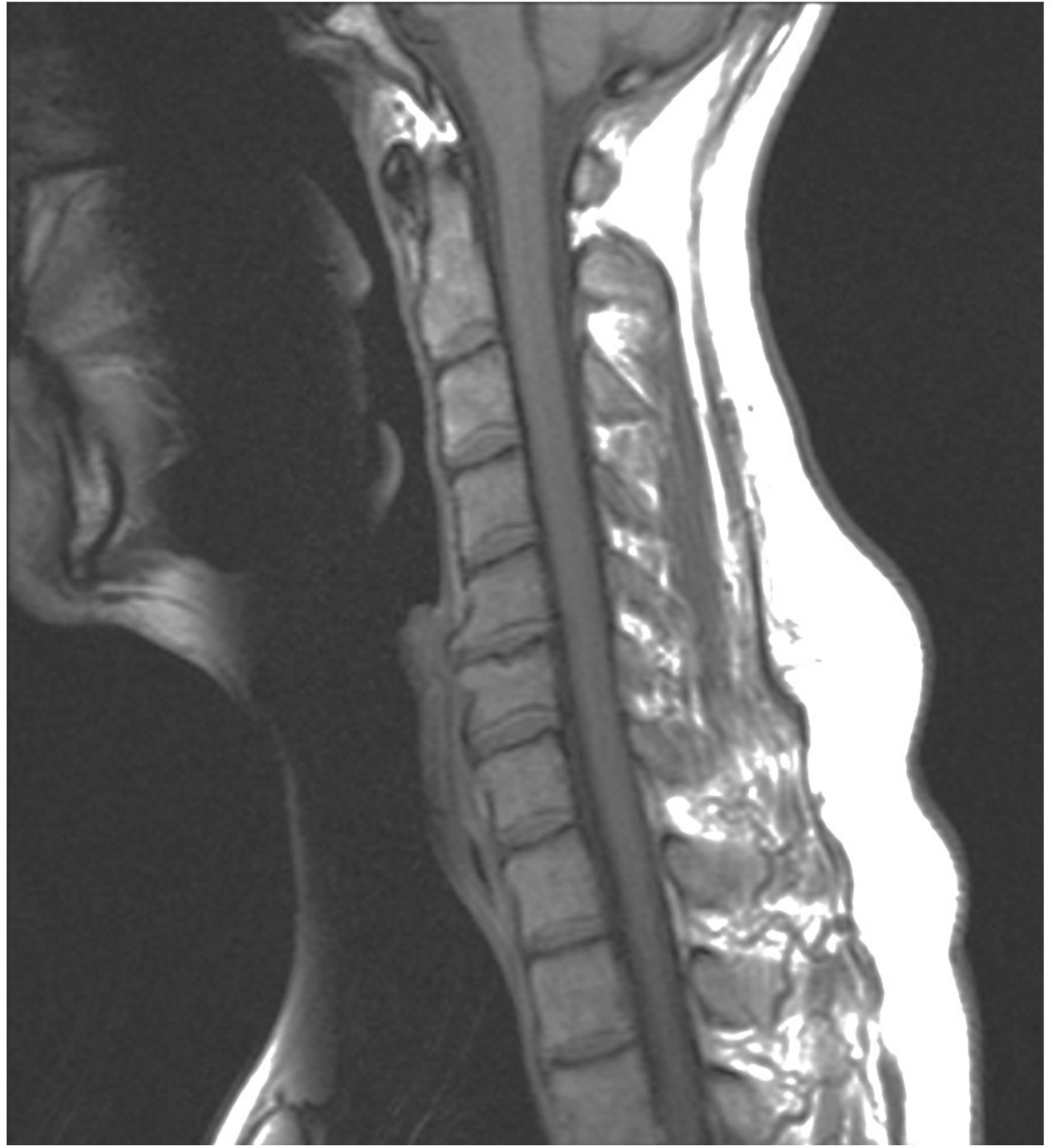
MRI

- Examples:
Cardiac MRI
 - Cardiovascular disease



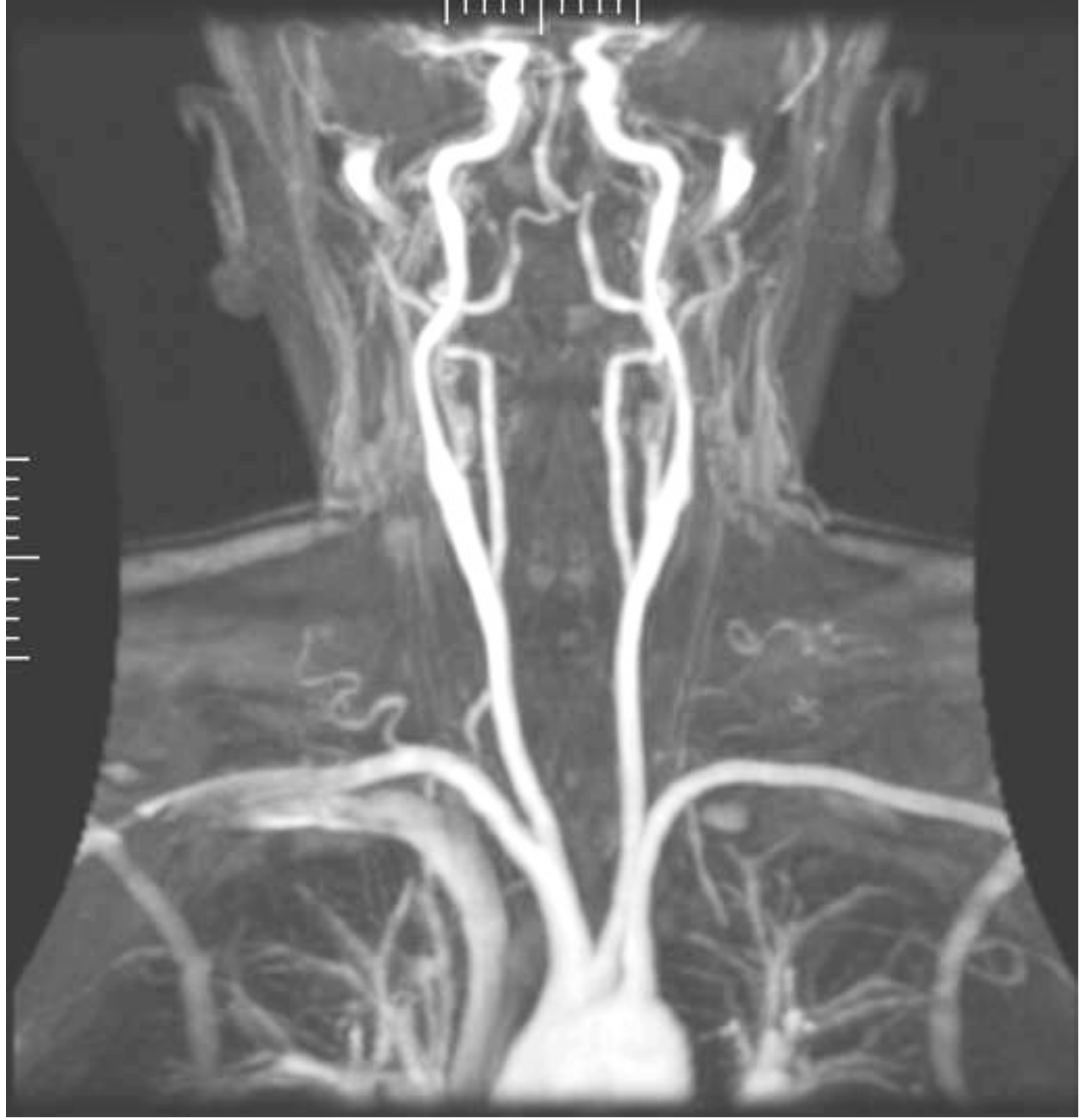
MRI

- Examples:
Spinal MRI



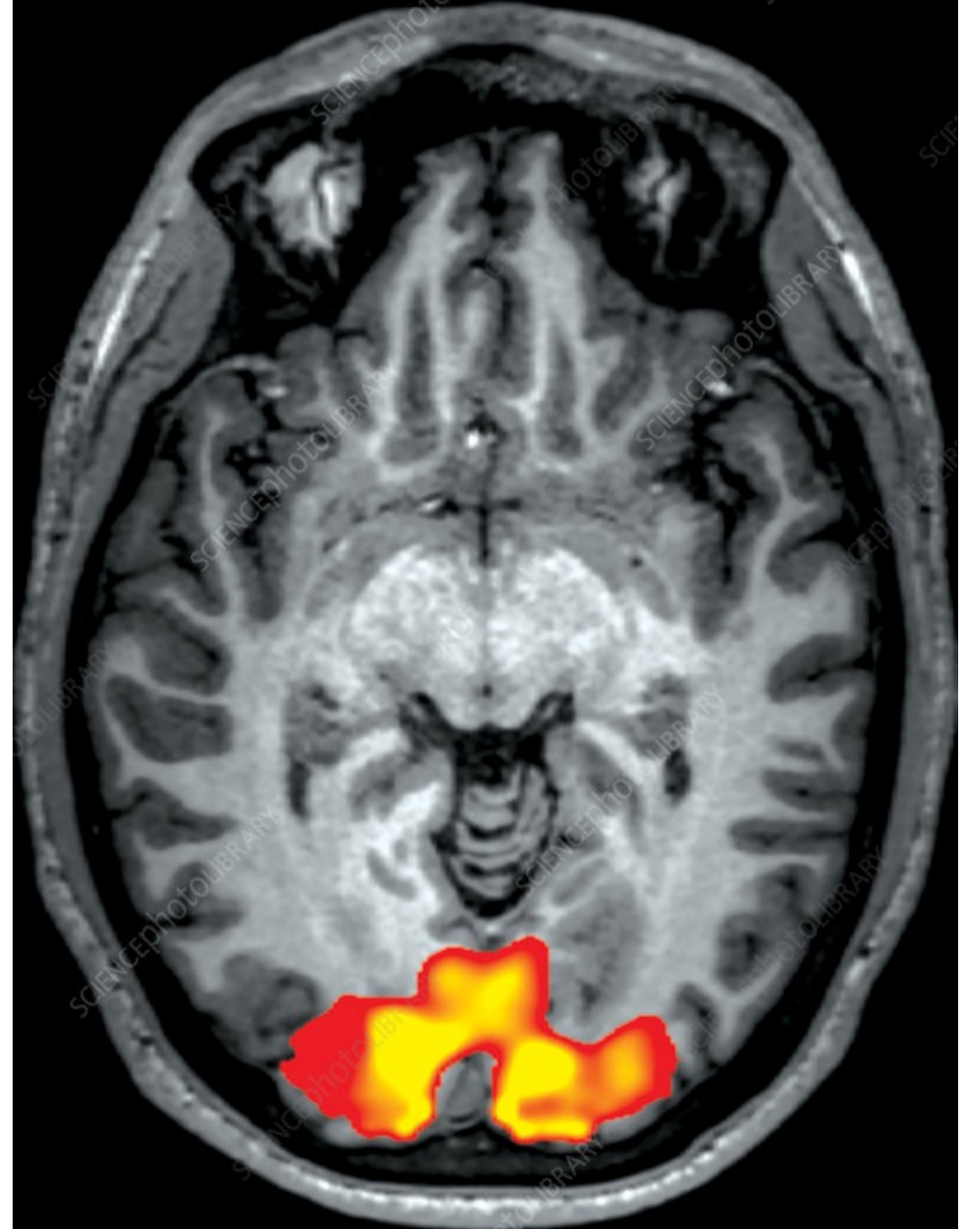
MRI

- Examples:
MR Angiography
 - Artery blockages



MRI

- Examples:
Functional MRI
 - Activation in brain regions
 - Study diseased or injured brain

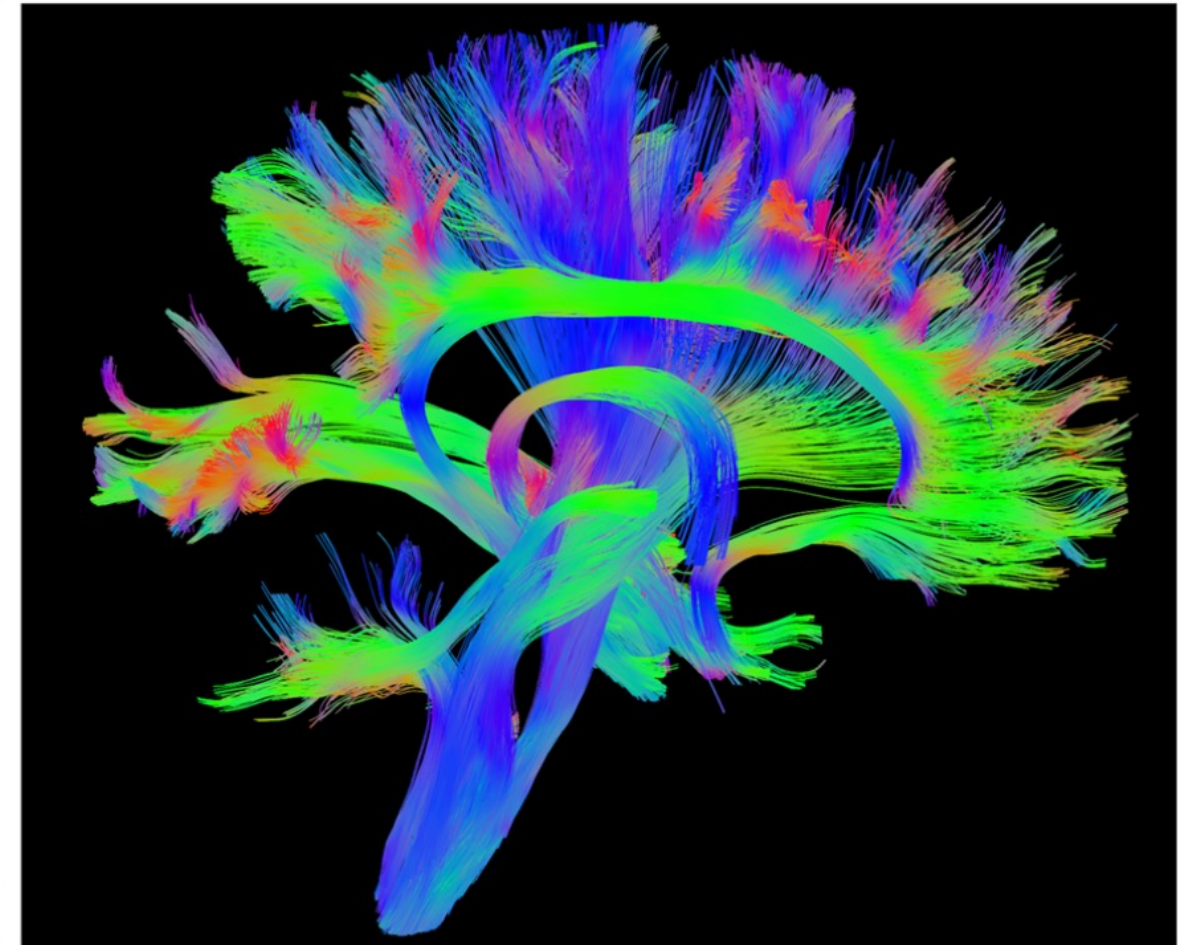


- Examples: Diffusion MRI

International Journal of Computer Vision (2021) 129:656–680



(a) The cerebral white matter (Williams et al., 1997)



(b) The ISMRM15 tractogram (Maier-Hein et al., 2017)

Fig. 1 Juxtaposition of (a) an anatomical sample of the cerebral white matter illustrating the fibrous white matter bundles, and (b) a reconstruction of white matter pathways by tractography from a DWI

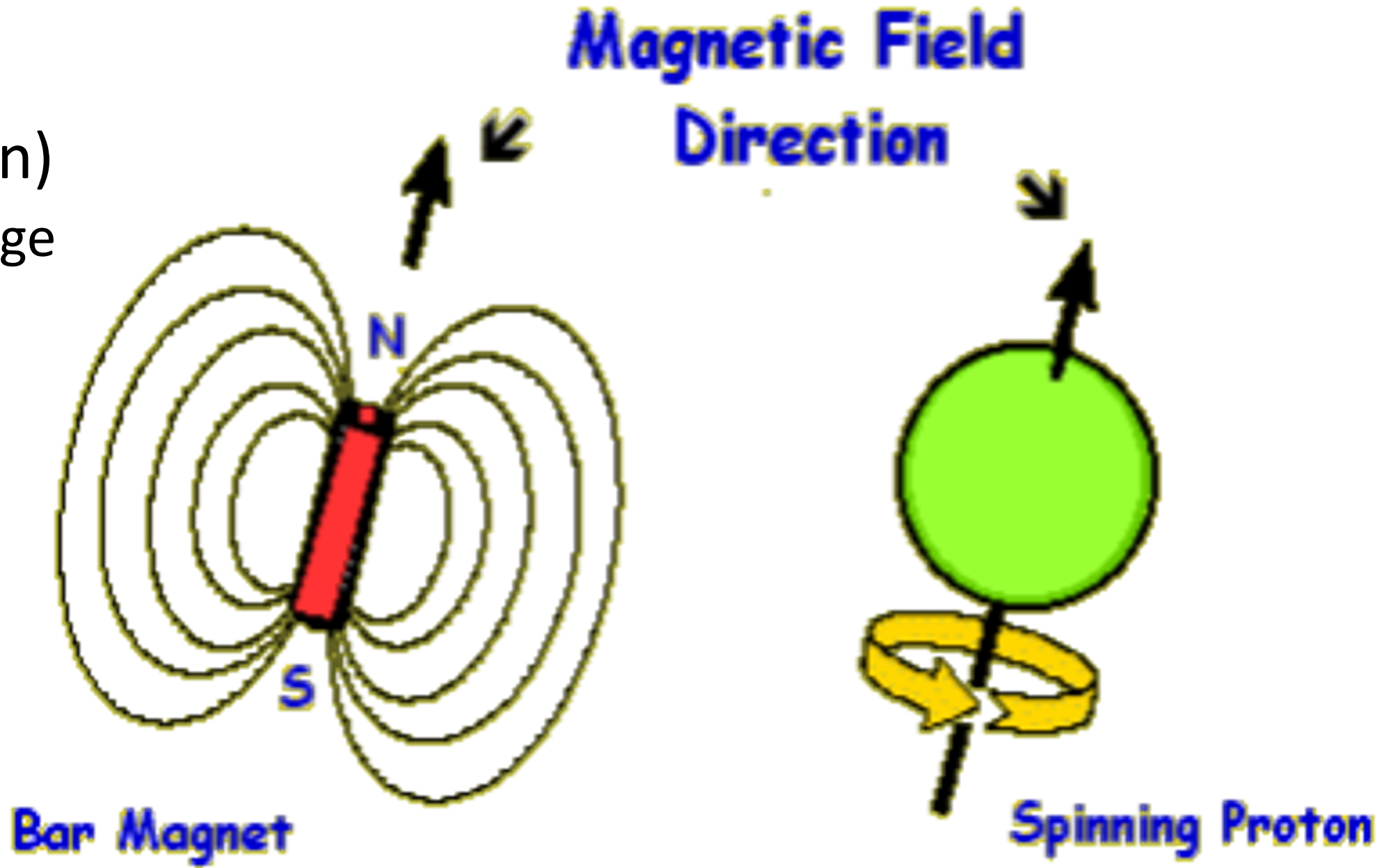
measurement. The RGB colors correspond to the projection of the local xyz fiber orientation ($R = x$, $G = y$, $B = z$)

MRI

- MRI advantages
 - No ionizing radiation (unlike Xray, CT, PET)
 - Better details within soft tissues in human body
 - Versus CT (bone)
- MRI disadvantages
 - Slower than Xray. Advanced MRI can be slower than CT.
 - Xray: fraction of a second
 - CT: around few minutes
 - MRI: few seconds to 10-20-or-more minutes
 - Machine is more expensive
 - Machine operation and maintenance needs more skill
 - Interferes with metallic objects within patients
 - Cardiac pacemaker, ear implant, dental crown

MRI

- Spinning proton (H^+ ion)
 - A moving electric charge acts like a magnet (magnetic dipole)
 - H^+ ions in water, and some in fats



- **Population** of spinning protons

- Assume non-magnetic body tissue, in absence of external magnetic field
- Dipoles oriented randomly (uniformly) along directions
- **Aggregate** magnetization = 0

MRI

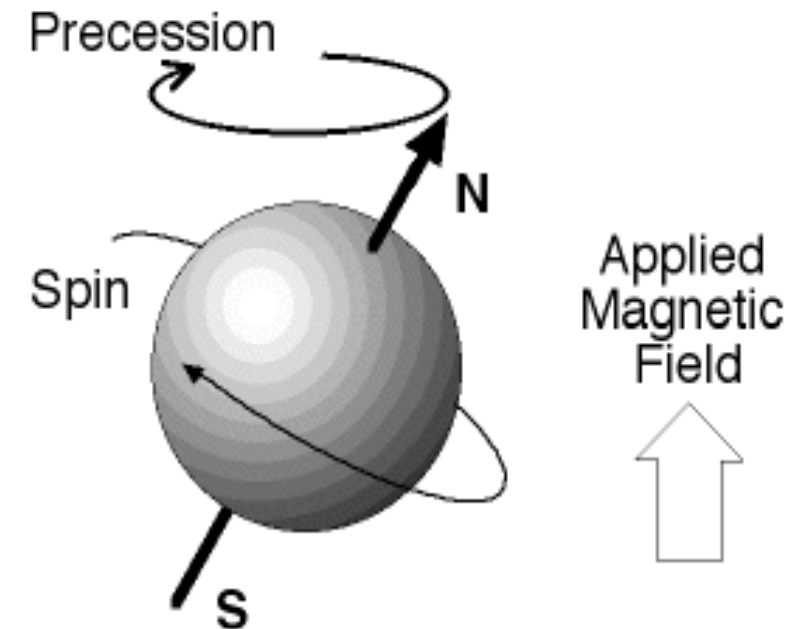
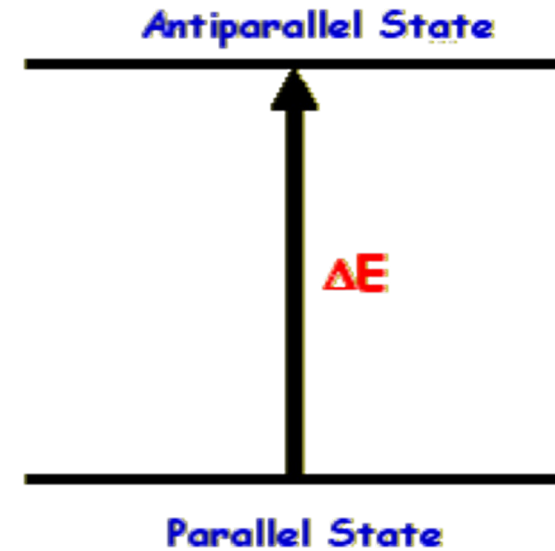
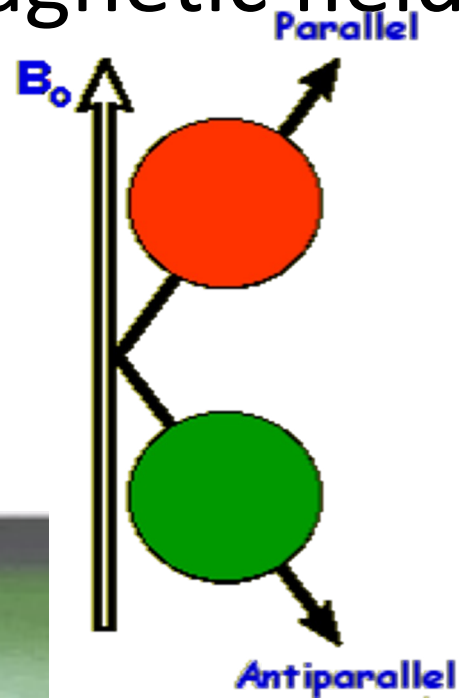
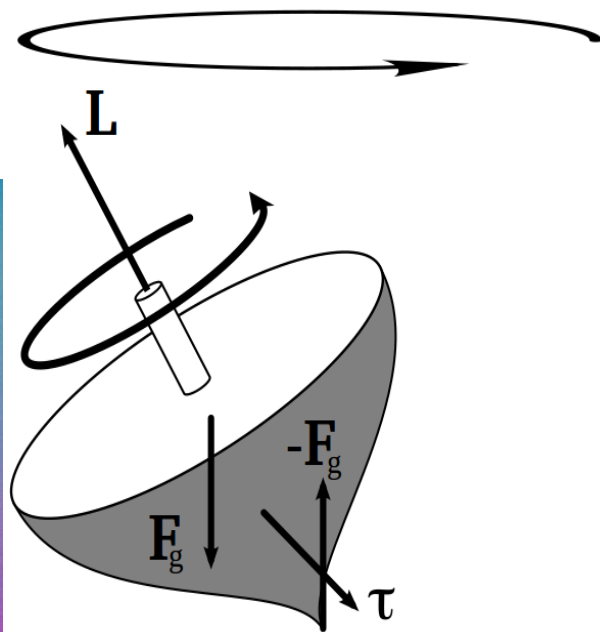
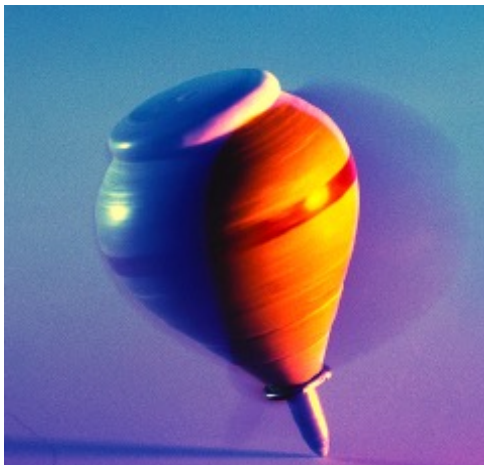
- **Single** spinning proton in a static **external** magnetic field

1) Starts **aligning** itself **along** or opposite **field**

- Alignment along field is lowest energy state

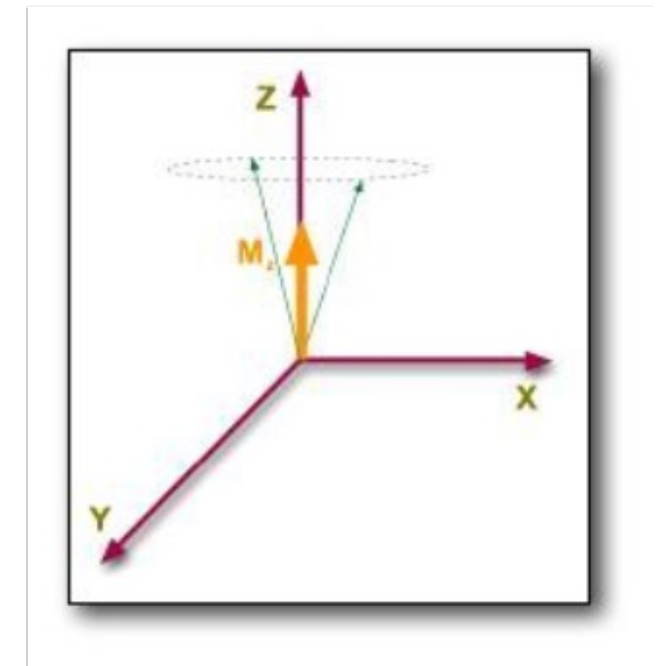
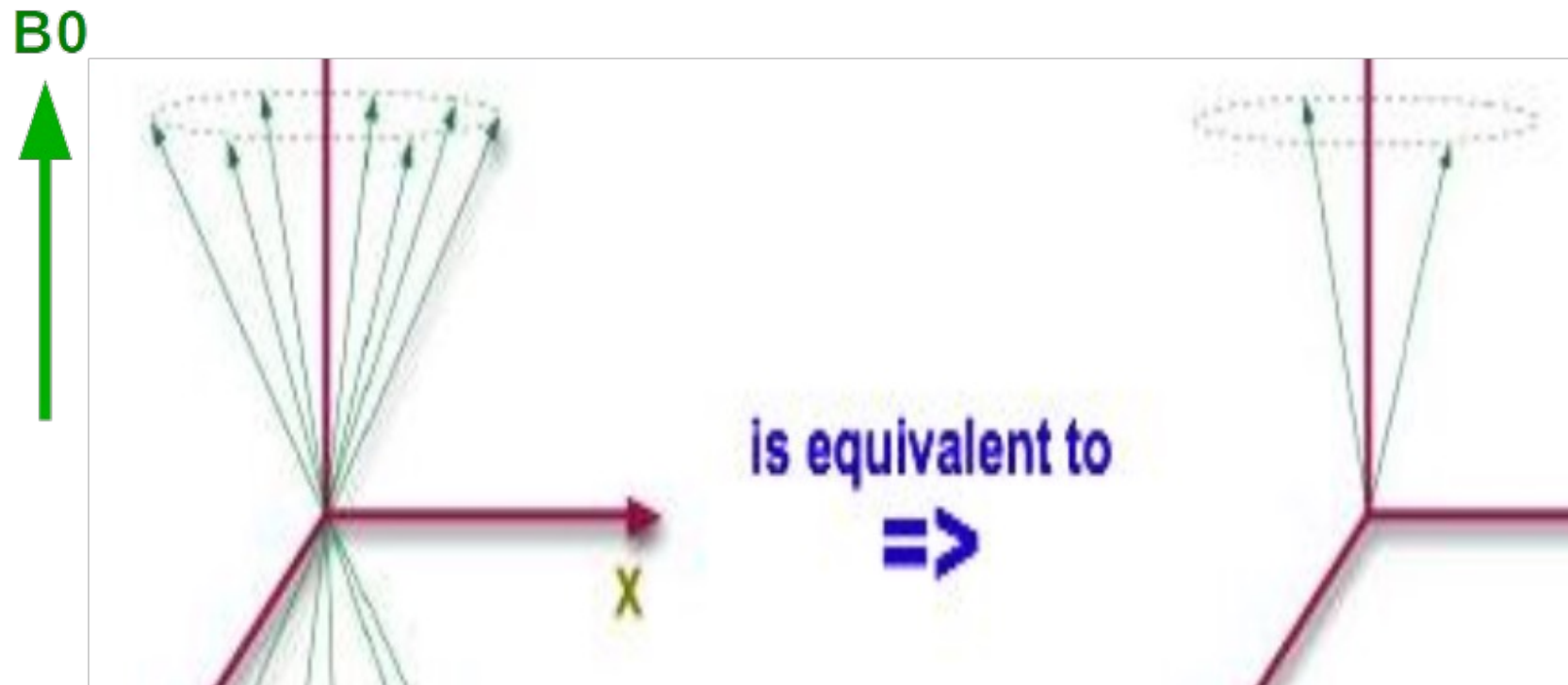
2) **Precession**, while spinning

- Spins/rotates around a central axis
- Central axis itself rotates around axis of external field



MRI

- **Population** of spinning protons in a static external magnetic field
- At **equilibrium**, spins become “**polarized**”
 - Protons precess about +Z or -Z axes
 - More protons precess around +Z axis, than -Z axis
 - Phase distribution across precessing protons is uniform
- **Aggregate magnetization (M)** along static field (B_0)



Nuclear Induction

F. BLOCH

Stanford University, California

(Received July 19, 1946)

- Bloch Equations
- Model what ?
- Process for aggregate magnetization, within a specific tissue, to reach equilibrium in (general) time-varying external magnetic field
 - 1) Rate of change of aggregate magnetization
 - 2) Precession of aggregate magnetization
 - 3) Longitudinal and transverse “relaxation”
 - Convention: longitudinal axis is +Z axis; transverse plane is XY plane
 - 4) Equilibrium state
 - “Equations of motion of nuclear magnetization”
- Phenomenological: Consistent with fundamental theory, but not (yet) derived from fundamental theory in classical/quantum physics

MRI

- Bloch Equations
- B_0 := a strong uniform steady magnetic field; oriented along +Z axis
 - Vector in 3D physical space
 - $B_0 := \langle 0, 0, b_0 \rangle$, with b_0 positive
- Now, consider additional (time-varying) magnetic fields introduced
- $B(t)$:= **total external magnetic field** experienced by sample at time t
 - Vector in 3D physical space
 - $B(t) := \langle b_x(t), b_y(t), b_z(t) \rangle$
- $M(t)$:= **aggregate magnetic moment** of nuclei in sample at time t
 - Vector in 3D physical space
 - $M(t) := \langle m_x(t), m_y(t), m_z(t) \rangle$

MRI

- Bloch Equations:
$$\frac{dM}{dt} = \gamma M \times B - \frac{\langle m_x, m_y, 0 \rangle}{T_2} - \frac{\langle 0, 0, m_z - m_{eq} \rangle}{T_1}$$
- t = time **where γ, T_1, T_2, m_{eq} are constants.**
- γ = gyromagnetic ratio (for H^+ protons in water, $\gamma \cong 43$ MHz per Tesla)
- T_1, T_2 depend on tissue properties
 - Tissue = magnetic nuclei contained within a specific biochemical environment
- m_{eq} = constant
 - Depends only on “proton” (i.e., H^+) density (PD)
= num. of protons in unit vol. of tissue / num. of protons in same vol. of water
(expressed as a percentage; PD for water is 100%; $m_{eq} \propto PD$)
 - Comes out as aggregate magnetic moment of nuclei in tissue at equilibrium
(as $t \rightarrow \infty$) when exposed to a static magnetic field

MRI

- T_1 , T_2 depend on tissue properties

At a main field of 1.5 T

Tissue type	Approximate T_1 value in ms	Approximate T_2 value in ms
Adipose tissues	240-250	60-80
Whole blood (deoxygenated)	1350	50
Whole blood (oxygenated)	1350	200
Cerebrospinal fluid (similar to pure water)	4200 - 4500	2100-2300
Gray matter of cerebrum	920	100
White matter of cerebrum	780	90
Liver	490	40
Kidneys	650	60-75
Muscles	860-900	50

MRI

- Bloch Equations
in
Static Magnetic Field
 $B_0 := \langle 0, 0, b_0 \rangle$
 - Independent of time and location
 - b_0 is 1.5-3 tesla
- Substitute $B(t) = B_0$
- Then, Bloch equations reduce to:

$$\frac{dM}{dt} = \gamma M \times B - \frac{\langle m_x, m_y, 0 \rangle}{T_2} - \frac{\langle 0, 0, m_z - m_{eq} \rangle}{T_1}$$

where γ, T_1, T_2, m_{eq} are constants.

$$\frac{dm_x}{dt} = \gamma b_0 m_y(t) - \frac{m_x(t)}{T_2}$$

$$\frac{dm_y}{dt} = -\gamma b_0 m_x(t) - \frac{m_y(t)}{T_2}$$

$$\frac{dm_z}{dt} = -\frac{m_z(t) - m_{eq}}{T_1}$$

MRI

- Bloch Equations in **Static** Magnetic Field
- Solution of these equations is:

$$\begin{aligned}m_x(t) &= e^{-t/T_2} \left(m_x(0) \cos(w_0 t) - m_y(0) \sin(w_0 t) \right) \\m_y(t) &= e^{-t/T_2} \left(m_x(0) \sin(w_0 t) + m_y(0) \cos(w_0 t) \right) \\m_z(t) &= m_z(0) e^{-t/T_1} + m_{eq} \left(1 - e^{-t/T_1} \right)\end{aligned}$$

$$\begin{aligned}\frac{dm_x}{dt} &= \gamma b_0 m_y(t) - \frac{m_x(t)}{T_2} \\ \frac{dm_y}{dt} &= -\gamma b_0 m_x(t) - \frac{m_y(t)}{T_2} \\ \frac{dm_z}{dt} &= -\frac{m_z(t) - m_{eq}}{T_1}\end{aligned}$$

where $m_{eq} := m_z(t = \infty)$ and $w_0 := -\gamma b_0$ is called **Larmor frequency**

- At **$b_0 = 3$ Tesla**, for **H^+ protons in water**, Larmor frequency **$w_0 = 129$ MHz**
 - Angular frequency: $+1 \text{ Hz} = +2\pi \text{ radians / sec}$

MRI

- Bloch Equations in Static Magnetic Field B_0

$$m_x(t) = e^{-t/T_2} \left(m_x(0) \cos(\omega_0 t) - m_y(0) \sin(\omega_0 t) \right)$$

$$m_y(t) = e^{-t/T_2} \left(m_x(0) \sin(\omega_0 t) + m_y(0) \cos(\omega_0 t) \right)$$

$$m_z(t) = m_z(0)e^{-t/T_1} + m_{eq} \left(1 - e^{-t/T_1} \right)$$

- Initial condition $M(t=0)$
- During a small time interval $[t, t+\Delta t]$, aggregate magnetization $M(t)$:
 - Has zero/non-zero components along Z axis, or within XY plane, as per $M(t=0)$
 - Is **precessing** along +Z axis:
 - $m_z(t)$: magnitude and sign stays same (Δt much smaller than T_1)
 - $m_{xy}(t)$ magnitude: stays same (Δt much smaller than T_2)
 - $m_{xy}(t)$ direction: rotates around +Z ; large frequency ω_0 implies $(\omega_0 \Delta t)$ is significant
 - Note: individual nuclei precess around aggregate magnetization vector M
- What happens to $M(t)$ at equilibrium ($t \rightarrow \infty$) ?
 - Aggregate magnetization has magnitude m_{eq} aligned along B_0

MRI

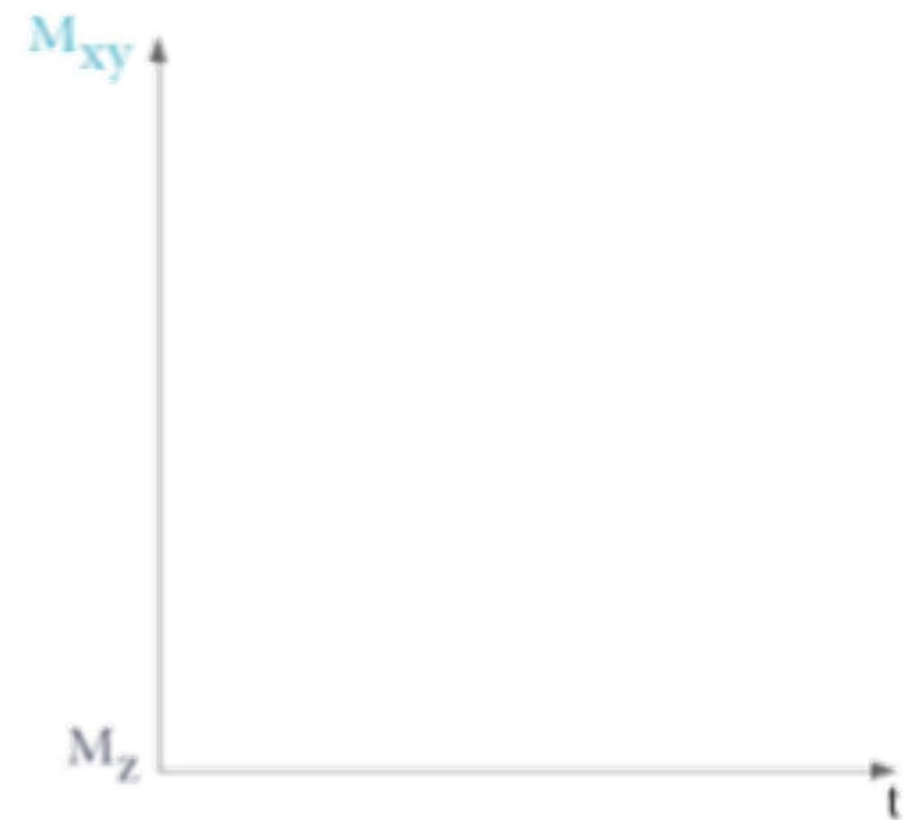
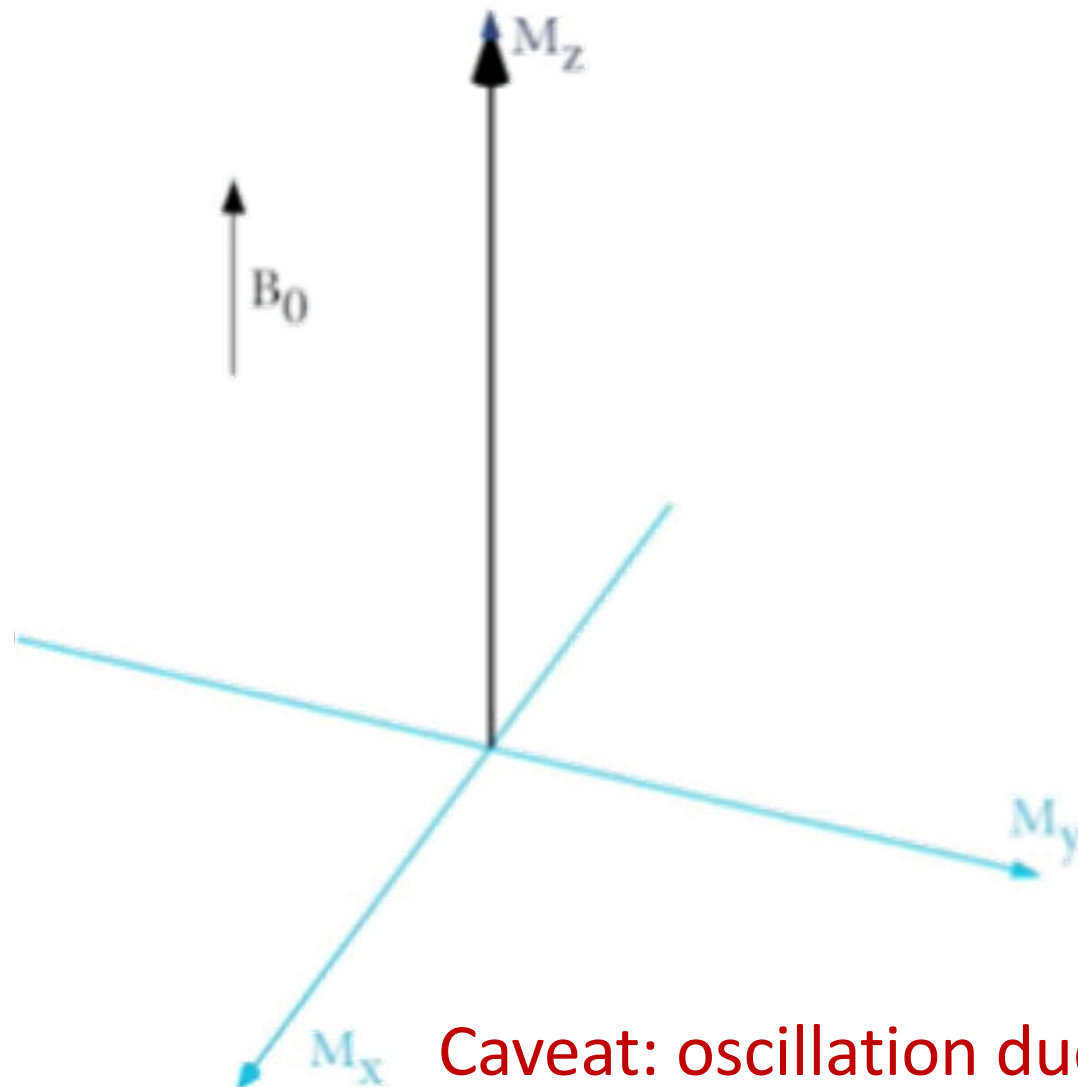
- Bloch Equations in a **static** magnetic field B_0
- **Longitudinal (Z) relaxation**, T_1 dependent
- **Transverse (XY) relaxation**, T_2 dependent

- Graphs
assume
 $m_z(0) = 0,$
 $\| m_{xy}(0) \| > 0$

$$m_x(t) = e^{-t/T_2} \left(m_x(0) \cos(\omega_0 t) - m_y(0) \sin(\omega_0 t) \right)$$

$$m_y(t) = e^{-t/T_2} \left(m_x(0) \sin(\omega_0 t) + m_y(0) \cos(\omega_0 t) \right)$$

$$m_z(t) = m_z(0)e^{-t/T_1} + m_{eq} \left(1 - e^{-t/T_1} \right)$$



Caveat: oscillation due to precession much slower than shown

MRI

- Bloch Equations in a static field B_0
- T1-weighted MRI (**Saturation Recovery**)
 - Consider two tissues (each with same proton density)
 - For tissue type A, let T_1 value be T_{1A}
 - For tissue type B, let T_1 value be T_{1B}
 - Initial state of tissue
 - NO magnetization of tissue, $m_x(0) = m_y(0) = m_z(0) = 0$; called “saturation” state
 - Place tissue A in magnetic field B_0 . **After time t , measure $m_{zA}(t)$.**
 - Repeat experiment with tissue B. Measure $m_{zB}(t)$.
 - Question: What value of t will give maximum difference between magnetization of tissues A and B ?

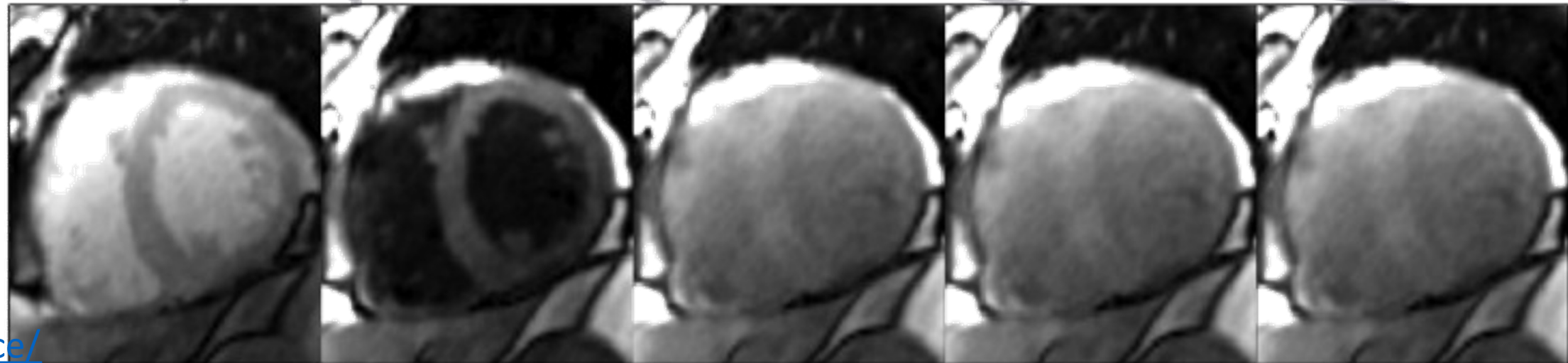
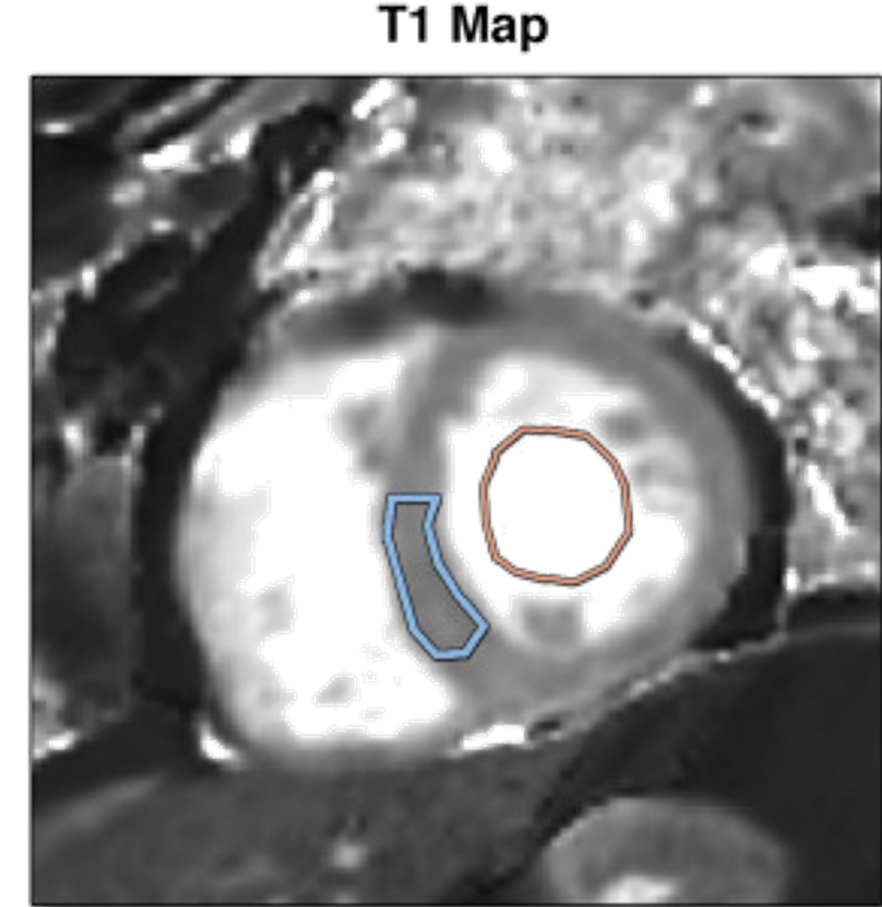
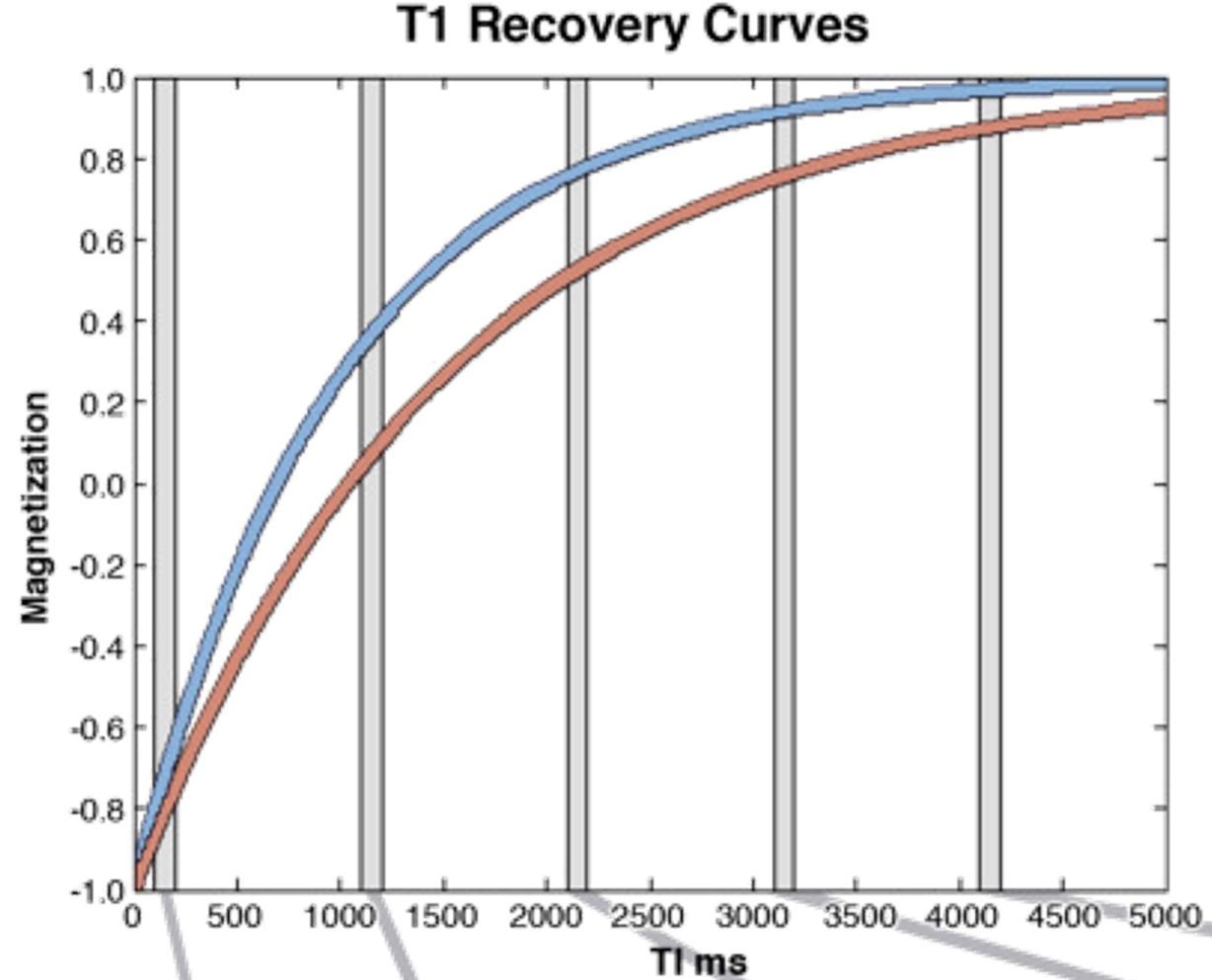
$$\arg \max_t m_{zA}(t) - m_{zB}(t)$$

$$t = \frac{\log(T_{1B}/T_{1A})}{1/T_{1A} - 1/T_{1B}}$$

$$m_x(t) = e^{-t/T_2} \left(m_x(0) \cos(w_0 t) - m_y(0) \sin(w_0 t) \right)$$
$$m_y(t) = e^{-t/T_2} \left(m_x(0) \sin(w_0 t) + m_y(0) \cos(w_0 t) \right)$$
$$m_z(t) = m_z(0)e^{-t/T_1} + m_{eq} \left(1 - e^{-t/T_1} \right)$$

MRI

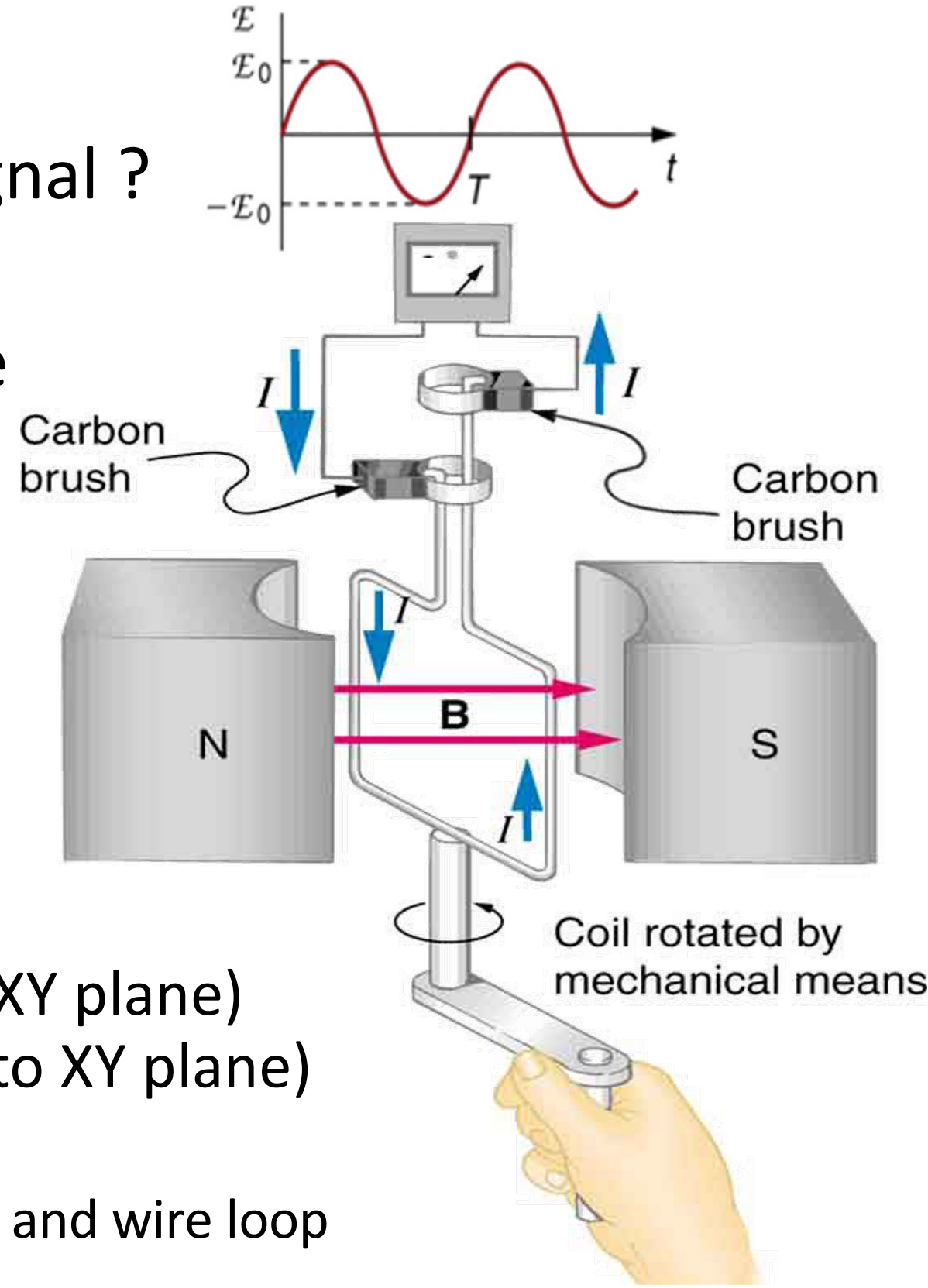
- Bloch Equations in a static magnetic field B_0
 - T_1 mapping
 - Multiple measurements over time
 - Fit curve



T1-Weighted Source Images

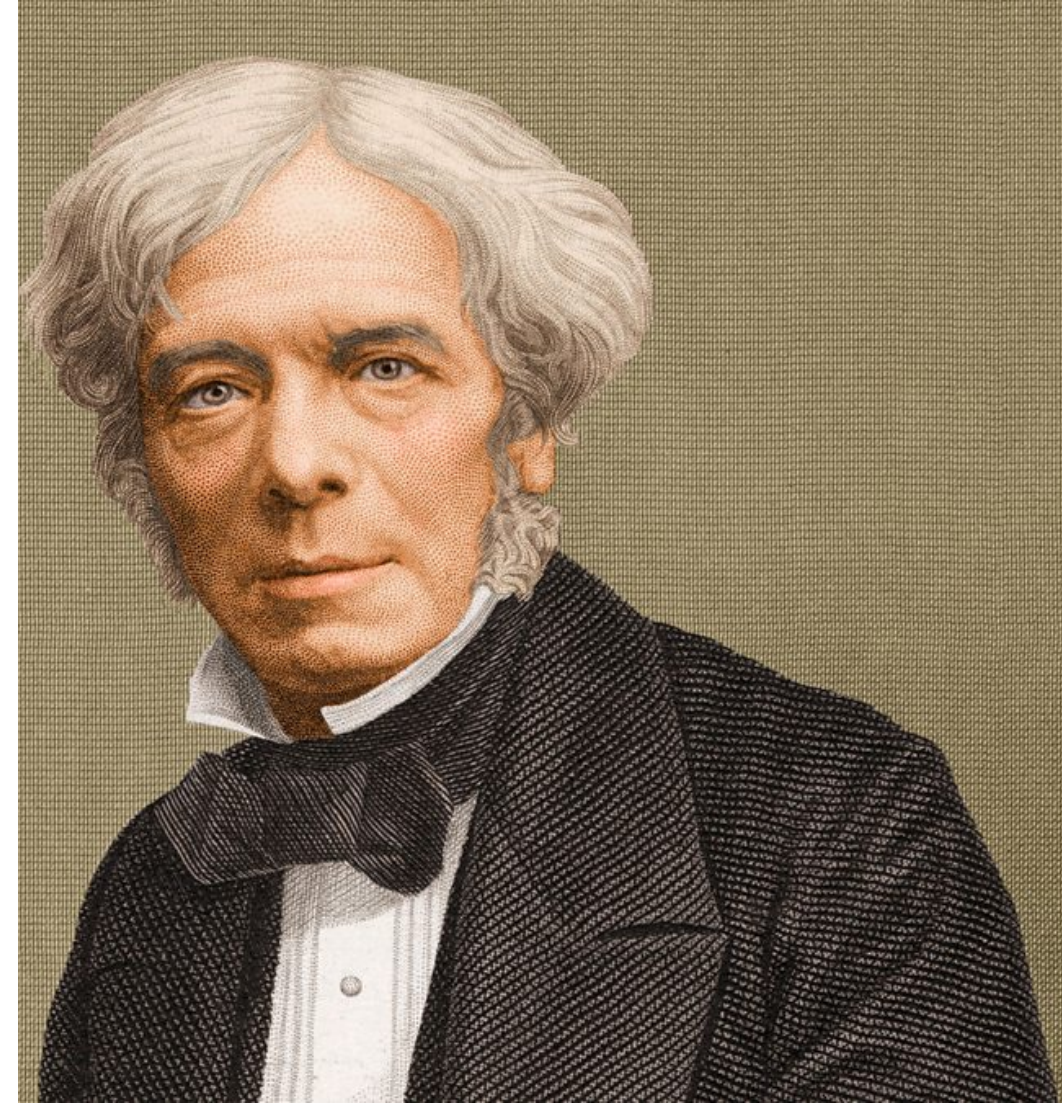
MRI

- But first, how do we even measure any signal ?
(even without any spatial localization)
- For T1-w MRI, how to **measure** magnitude of aggregate magnetization **$m_z(t)$** that is a (known) function of T_1 ?
 - $m_x(t)$, $m_y(t)$ aren't functions of T_1
 - Solution
 - Design $m_x(t)$, $m_y(t)$ to be 0
 - During $[t, t+\Delta t]$, somehow **make vector $m_z(t)$ rotate in XY plane**
- Faraday's Law: A changing magnetic field (in XY plane) induces a current in a wire loop (orthogonal to XY plane)
 - Same principle as of electrical generator
 - Only relative motion matters, between magnetic and wire loop

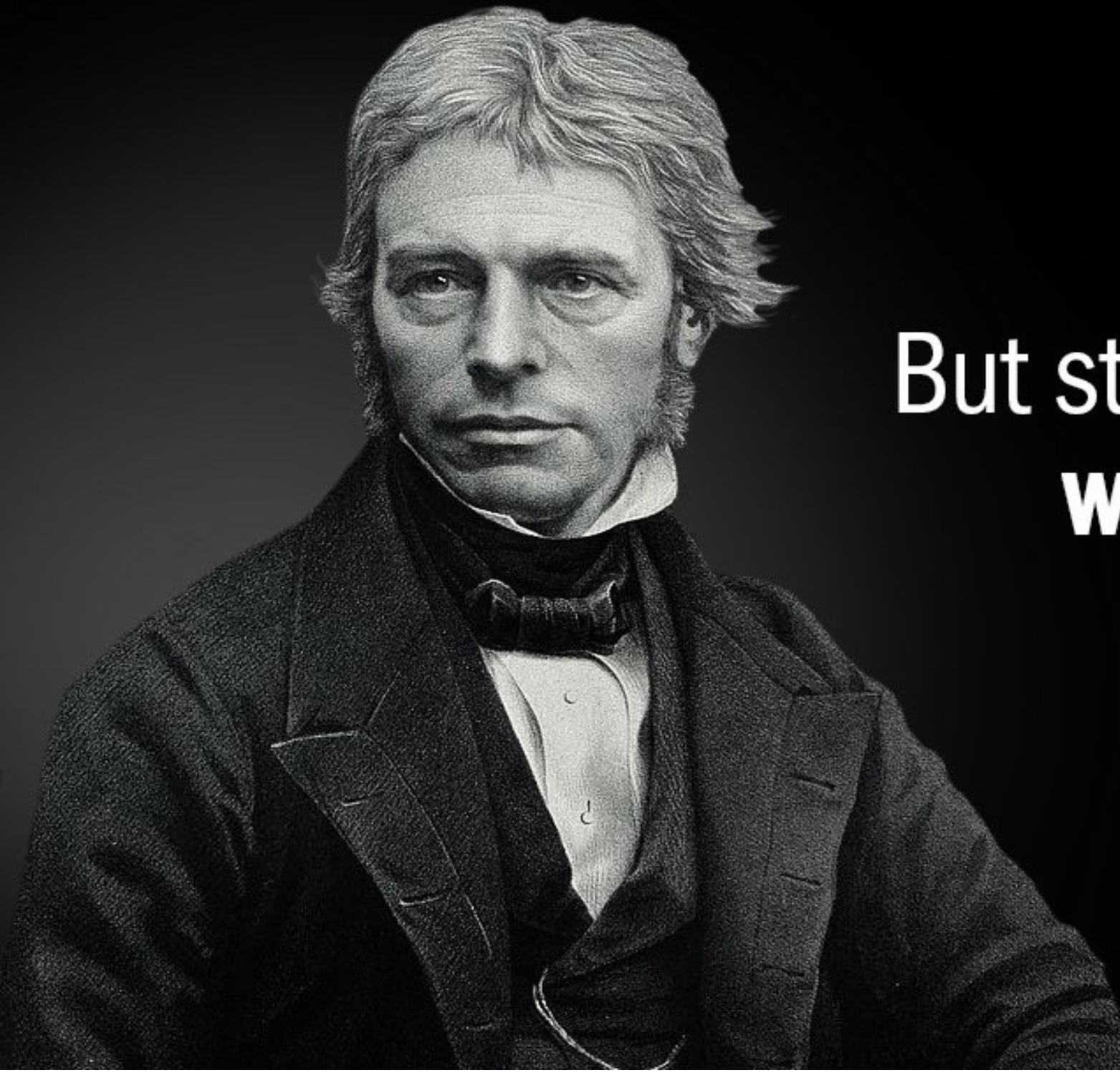


Michael Faraday

- Little formal education, educated himself
- Electromagnetic: Induction
- Electrochemistry: Electrolysis
- Chemistry: Discovered benzene
- Known for: Faraday's law of induction, Faraday effect, Faraday cage, Faraday cup, Faraday constant, Faraday paradox, Faraday's laws of electrolysis, Faraday rotator, Faraday-efficiency effect, Faraday wave, Faraday wheel, Lines of force
- "... there is no honour too great to pay to the memory of Faraday, one of the greatest scientific discoverers of all time"
– Ernest Rutherford



Michael Faraday



**But still try, for who knows
what is possible?**

– Michael Faraday

Michael Faraday



Michael Faraday

It is on record that when a young aspirant asked Faraday the secret of his success as a scientific investigator, he replied, 'The secret is comprised in three words- Work, Finish, Publish.'



Lectures which **really teach**
will **never be popular**;
lectures which are **popular**
will **never really teach.**

– *Michael Faraday*

MRI

- RF magnetic-field pulse $B_1(t)$ **and** Resonance
- (1) Take a system precessing around B_0 and
(2) Apply an oscillating magnetic field in XY plane
- Static magnetic field: $B_0 := \langle 0, 0, b_0 \rangle$
- Time-varying external oscillating magnetic field in XY plane:
 $B_1(t) := \langle b_1 \cos(\omega t), b_1 \sin(\omega t), 0 \rangle$
 - b_1 and ω are controllable parameters
 - b_1 isn't time-varying, but constant, typically around 30 micro-tesla
- Net magnetic field:
 $B(t) := B_0 + B_1(t) = \langle b_1 \cos(\omega t), b_1 \sin(\omega t), b_0 \rangle$

MRI

- RF magnetic-field pulse $B_1(t)$ and Resonance
- $B(t) := B_0 + B_1(t) = \langle b_1 \cos(\omega t), b_1 \sin(\omega t), b_0 \rangle$
- **Assumption:** Compared to T_1 or T_2 , our design will ensure that there will only be a **tiny duration** for which $B_1(t)$ will be non-zero

$$\frac{dM}{dt} = \gamma M \times B - \frac{\langle m_x, m_y, 0 \rangle}{T_2} - \frac{\langle 0, 0, m_z - m_{eq} \rangle}{T_1}$$

where γ, T_1, T_2, m_{eq} are constants.

- RF “pulse”
- Ignore terms containing T_1, T_2 in Bloch equation
- Simplified Bloch equations $\rightarrow$

$$\frac{dm_x}{dt} = \gamma b_0 m_y - \gamma b_1 m_z \sin(\omega t)$$

$$\frac{dm_y}{dt} = -\gamma b_0 m_x + \gamma b_1 m_z \cos(\omega t)$$

$$\frac{dm_z}{dt} = \gamma b_1 m_x \sin(\omega t) - \gamma b_1 m_y \cos(\omega t)$$

MRI

- RF magnetic-field pulse $B_1(t)$ and Resonance
- Introduce rotating (orthonormal) coordinate frame $e_1 e_2$ in XY plane

$$e_1(t) := \langle \cos(\omega t), \sin(\omega t) \rangle$$

$$e_2(t) := \langle -\sin(\omega t), \cos(\omega t) \rangle$$

- We want to design/re-write: $\langle m_x(t), m_y(t) \rangle = u(t)e_1(t) + v(t)e_2(t)$
 - (column) vector $[u, v]^T$
gets rotated by matrix $[e_1^T \mid e_2^T]$
to give (column) vector $[m_x, m_y]^T$
 - This gives:
$$u(t) := m_x(t) \cos(\omega t) + m_y(t) \sin(\omega t)$$
$$v(t) := m_y(t) \cos(\omega t) - m_x(t) \sin(\omega t)$$

MRI

- RF magnetic-field pulse $B_1(t)$ and Resonance

- Rewrite **simplified Bloch equations in rotating coordinate frame**

- Differentiate **LHS and RHS** w.r.t. 't'

- In simplified Bloch equations ...

- ... substitute derivatives to give:

$$u(t) := m_x(t) \cos(\omega t) + m_y(t) \sin(\omega t)$$

$$v(t) := m_y(t) \cos(\omega t) - m_x(t) \sin(\omega t)$$

$$\frac{dm_x}{dt} = \gamma b_0 m_y - \gamma b_1 m_z \sin(\omega t)$$

$$\frac{dm_y}{dt} = -\gamma b_0 m_x + \gamma b_1 m_z \cos(\omega t)$$

$$\frac{dm_z}{dt} = \gamma b_1 m_x \sin(\omega t) - \gamma b_1 m_y \cos(\omega t)$$

$$\frac{du}{dt} = (\gamma b_0 + \omega)v$$

$$\frac{dv}{dt} = -(\gamma b_0 + \omega)u + \gamma b_1 m_z$$

$$\frac{dm_z}{dt} = -\gamma b_1 v$$

MRI

- RF magnetic-field pulse $B_1(t)$ and Resonance

- Resonance

- Design frequency of RF pulse
s.t. $\omega := \omega_0 = -\gamma b_0$
= Larmor frequency of
 H^+ protons in B_0

- Solution at resonance:

$$u(t) = u(0)$$

$$v(t) = v(0) \cos(-\gamma b_1 t) - m_z(0) \sin(-\gamma b_1 t)$$

$$m_z(t) = m_z(0) \cos(-\gamma b_1 t) + v(0) \sin(-\gamma b_1 t)$$

- Aggregate magnetization precesses around $e_1(t)$ axis
 - $e_1(t)$ axis (itself) rotating at Larmor frequency

$$\frac{du}{dt} = (\gamma b_0 + \omega)v$$

$$\frac{dv}{dt} = -(\gamma b_0 + \omega)u + \gamma b_1 m_z$$

$$\frac{dm_z}{dt} = -\gamma b_1 v$$

$$\frac{du}{dt} = 0$$

$$\frac{dv}{dt} = \gamma b_1 m_z$$

$$\frac{dm_z}{dt} = -\gamma b_1 v$$

MRI

• RF Pulse: 90-degree Pulse

1) Take a system precessing around B_0

- e.g., assume initial state is $m_x(0) = 0$; $m_y(0) = 0$; $m_z(0) > 0$
- Initial condition in rotated frame is: $u(0) = v(0) = 0$; because $m_x(0) = m_y(0) = 0$

2) Apply $B_1(t) := \langle b_1 \cos(\omega_0 t), b_1 \sin(\omega_0 t), 0 \rangle$ for specific duration τ_1

- γ 43 MHz, b_1 30 micro-tesla $\rightarrow \tau_1$ 1.2 ms $\tau_1 := \frac{1}{\gamma b_1} \frac{\pi}{2}$ where $\tau_1 \ll T_1, T_2$

• After time τ_1 : $u(\tau_1) = u(0) = 0$

$$v(\tau_1) = m_z(0)$$

$$m_z(\tau_1) = -v(0) = 0$$

- In rotating coordinate frame $(e_1(t), e_2(t), z)$, aggregate-magnetization vector changes from $\langle 0, 0, m_z(0) \rangle \rightarrow \langle 0, m_z(0), 0 \rangle$, i.e., 90-degree rotation
 - Magnetization flipped by 90 degrees

$$u(t) = u(0)$$

$$v(t) = v(0) \cos(-\gamma b_1 t) - m_z(0) \sin(-\gamma b_1 t)$$

$$m_z(t) = m_z(0) \cos(-\gamma b_1 t) + v(0) \sin(-\gamma b_1 t)$$

MRI

- RF Pulse: **90-degree Pulse used in Saturation Recovery**

1) Initial state: $m_x(0) = m_y(0) = m_z(0) = 0$

2) Let magnetization recover towards equilibrium

- At time t , aggregate magnetization is:

$$m_x(t) = m_y(t) = 0 \text{ and } m_z(t) = m_{eq}(1 - \exp(-t/T_1))$$

3) Apply 90-deg pulse. At time $t + \tau_1$:

$$u(t + \tau_1) = u(t) = 0$$

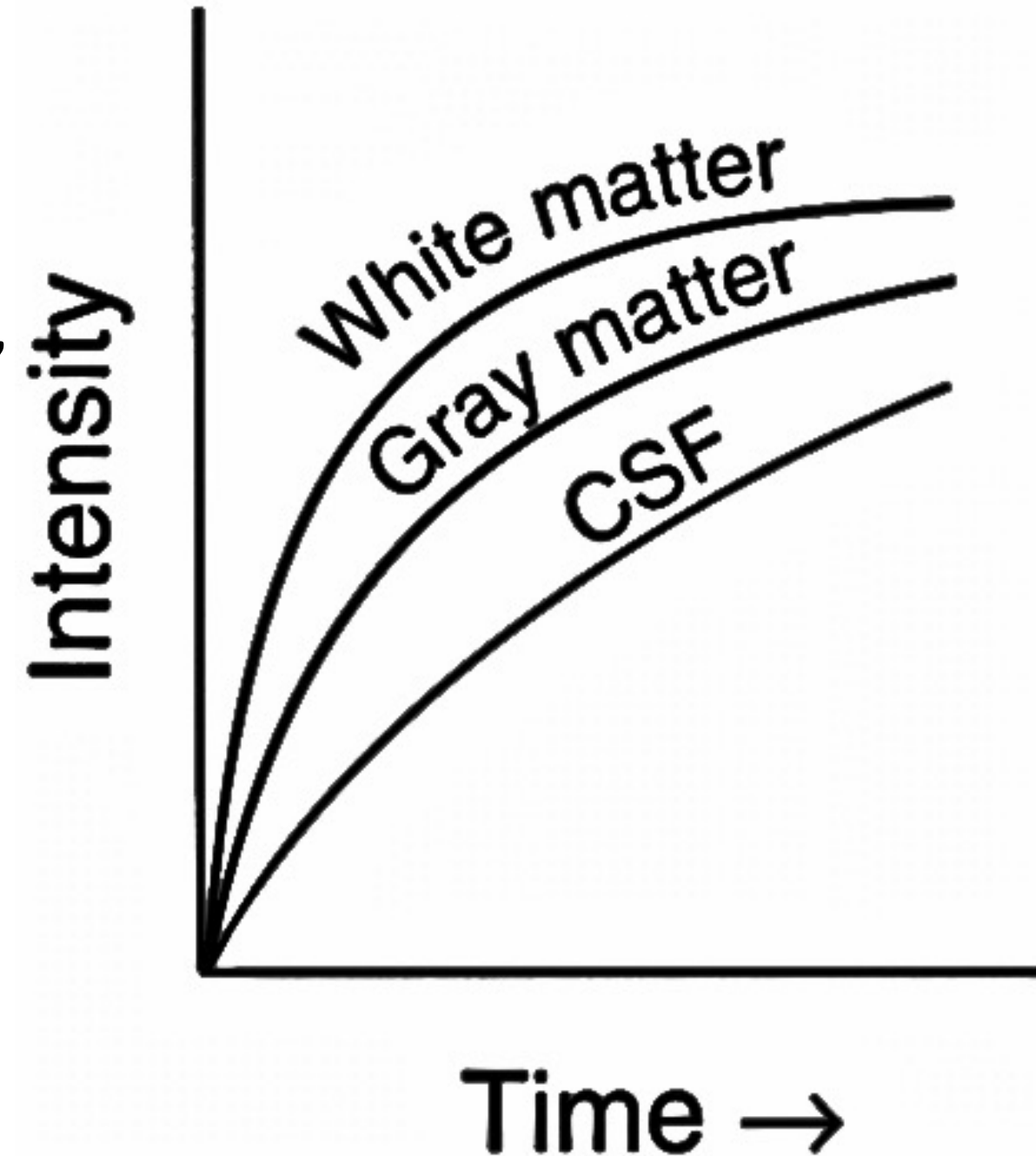
$$v(t + \tau_1) = m_z(t) = m_{eq}(1 - e^{-t/T_1})$$

$$m_z(t + \tau_1) = -v(t) = 0$$

4) Measure current $\propto m_{eq}(1 - \exp(-t/T_1))$

MRI

- Bloch Equations in a **static** magnetic field B_0
- If we have some way (we will see soon) to make **different measurements** at **different spatial locations** within tissue, then there could be an “imaging” application



MRI

- **Slice selection**

- Static magnetic field: $\mathbf{B}_0 := \langle 0, 0, b_0 \rangle$

- Time-varying field $\mathbf{B}_1(t) := \langle b_1(t) \cos(\omega t), b_1(t) \sin(\omega t), 0 \rangle$

 - Magnitude $b_1(t)$ is also time varying

- Spatially-varying “gradient” magnetic field

$$\mathbf{B}_G(\mathbf{p}) := \langle 0, 0, b_g(\mathbf{p}) \rangle$$

 - Spatial location ' $\mathbf{p}' = (x, y, z)$

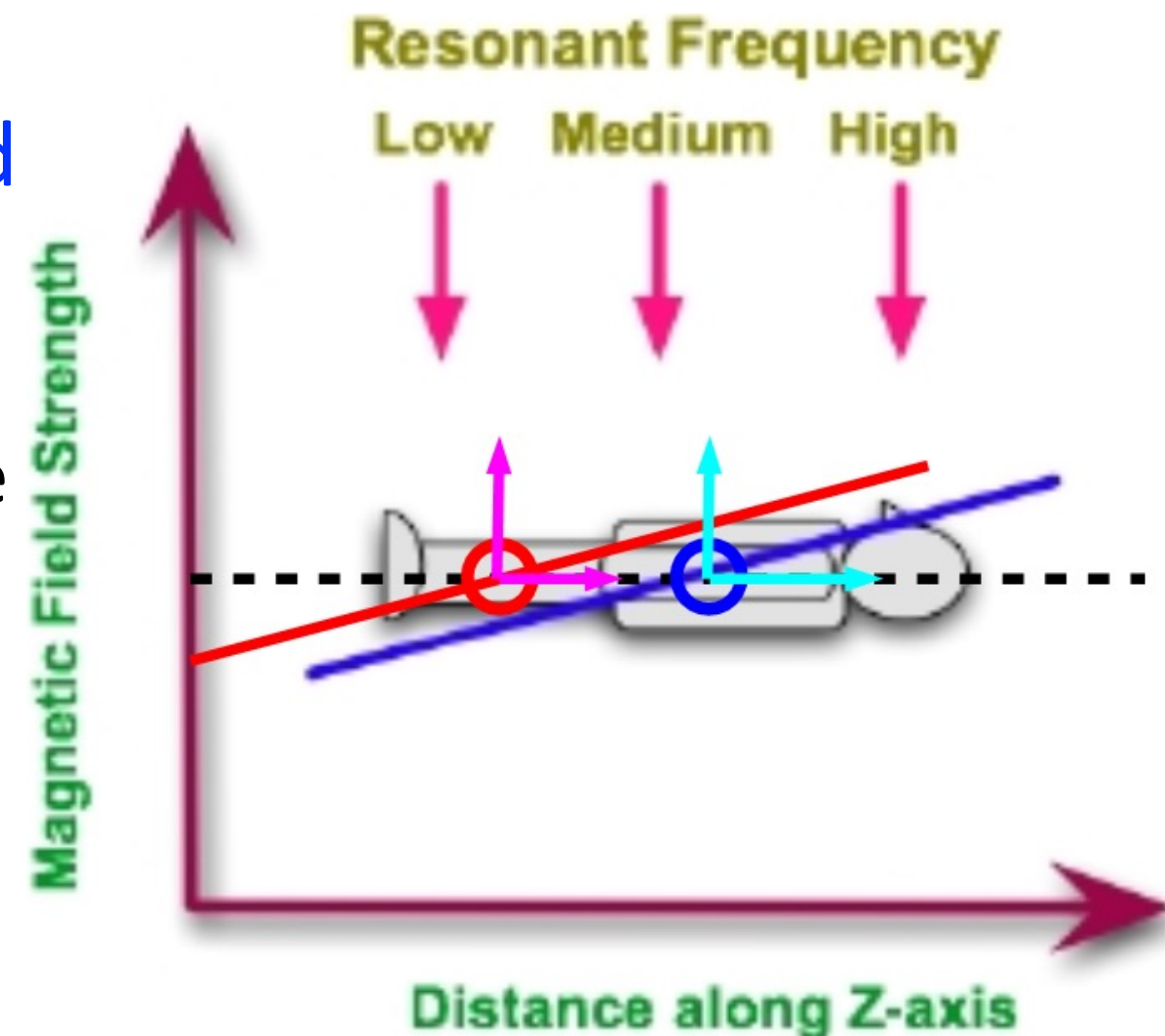
 - Design $b_g(\mathbf{p}) = gz + d$, can be positive/negative

 - Constants g, z . For now, assume $d = 0$.

- Net magnetic field is

$$\mathbf{B}(t, \mathbf{p}) := \mathbf{B}_0 + \mathbf{B}_G(\mathbf{p}) + \mathbf{B}_1(t)$$

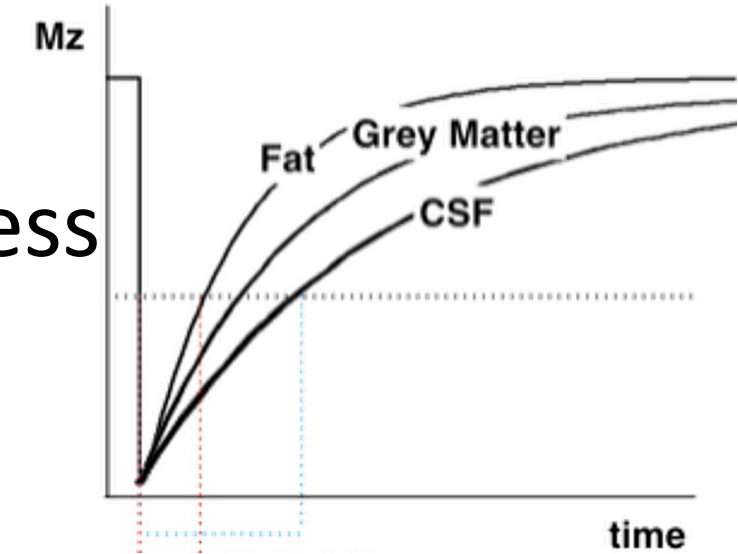
$$= \langle b_1(t) \cos(\omega t), b_1(t) \sin(\omega t), b_0 + b_g(\mathbf{p}) \rangle$$



MRI

- **Step 1:** In **static field B_0** , **saturation recovery** in progress

- At spatial location (x,y,z) , magnetization $M_z(x,y,z)$ depends on t and $T_1(x,y,z)$

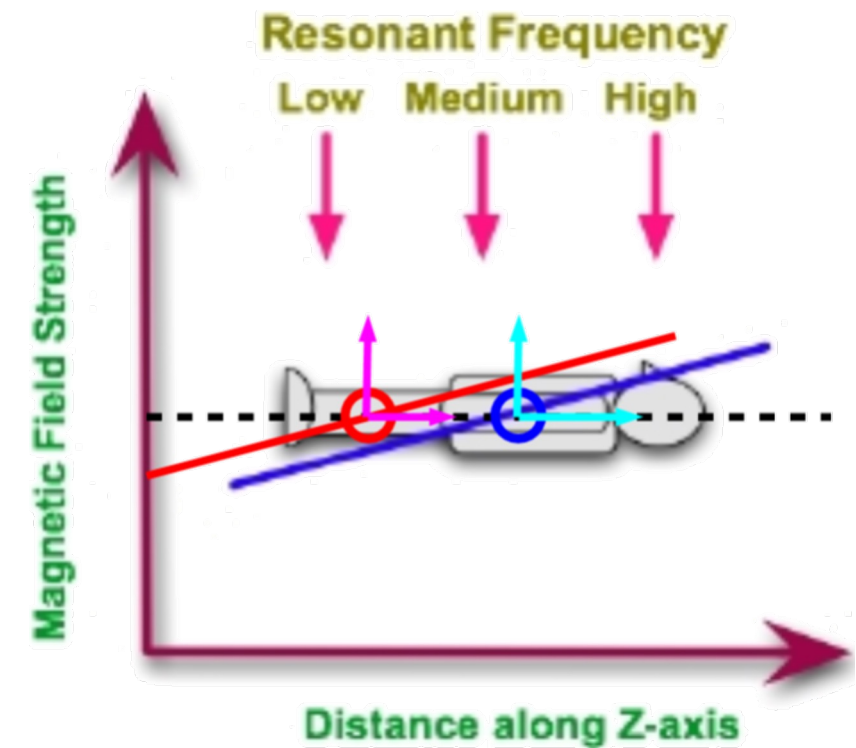


- **Step 2: Slice Selection**

- Simultaneously, turn on RF field $B_1(t')$ + Z-gradient field $B_G(p)$

- Turn off both

- Now, magnetization is **non-zero only in selected slice**; scaled $M_z(x,y,z)$ is **rotating within transverse XY plane** at **Larmour frequency $-\gamma b_0$**



- $M_z(x,y,z) = 0$ everywhere
- $M_{xy}(x,y,z) = 0$ for z outside selected slice
- $M_{xy}(x,y,z) \cong \gamma M_z(x,y,z; t, T_1(x,y,z))$
for z inside selected slice $[-\Delta z/2, +\Delta z/2]$
- All magnetization vectors rotate in sync/phase: $M_{xy}(x,y,z) \exp(i \omega_0 t)$ in complex domain

MRI

• Step 3: “Phase” Encoding

- Turn on Y-gradient field $\mathbf{G}_Y$

- Each Y ordinate is subject to different **static** magnetic field $\mathbf{B}_0 + \mathbf{B}_G(\mathbf{y}) = \langle 0, 0, b_0 + \mathbf{G}_Y \mathbf{y} \rangle$
- Each Y-ordinate's magnetization precesses at Y-dependent frequency

- Recall:

$$m_x(t) = e^{-t/T_2} \left(m_x(0) \cos(\omega_0 t) - m_y(0) \sin(\omega_0 t) \right)$$

$$m_y(t) = e^{-t/T_2} \left(m_x(0) \sin(\omega_0 t) + m_y(0) \cos(\omega_0 t) \right) \quad \omega_0 = -\gamma b_0$$

$$m_z(t) = m_z(0) e^{-t/T_1} + m_{eq} \left(1 - e^{-t/T_1} \right)$$

At $b_0 = 3$ Tesla, for H^+ protons in water,
Larmor frequency $\omega_0 = 129$ MHz

- Turn off Y-gradient field after time $t_Y \ll T_1, T_2$
- Each magnetization vector $\mathbf{N}_{xy}(x, y) \exp(i \omega_0 t)$ retains magnitude; develops phase lag proportional to y, t_Y
- Magnetization vector (rotating within XY plane):
 $\mathbf{N}_{xy}(x, y) \exp(i \omega_0 t) \exp(i \mathbf{G}_Y \mathbf{y} t_Y)$

MRI

• Step 4: “Frequency” Encoding

- Turn on **X**-gradient field $\mathbf{G}_x$

- Each **X** abscissa is subject to different **static** magnetic field $\mathbf{B}_0 + \mathbf{B}_G(\mathbf{x}) = \langle 0, 0, b_0 + \mathbf{G}_x \mathbf{x} \rangle$
- Each **X**-abscissa's magnetization precesses at **X-dependent frequency**

- Recall:

$$m_x(t) = e^{-t/T_2} \left(m_x(0) \cos(w_0 t) - m_y(0) \sin(w_0 t) \right)$$

$$m_y(t) = e^{-t/T_2} \left(m_x(0) \sin(w_0 t) + m_y(0) \cos(w_0 t) \right) \quad w_0 = -\gamma b_0$$

$$m_z(t) = m_z(0) e^{-t/T_1} + m_{eq} \left(1 - e^{-t/T_1} \right)$$

At $b_0 = 3$ Tesla, for H^+ protons in water,
Larmor frequency $w_0 = 129$ MHz

- After **time** $t_x \ll T_1, T_2$
- Each magnetization vector $\mathbf{N}_{xy}(x, y) \exp(i w_0 t) \exp(i G_y y t_y)$ retains magnitude; develops **phase lag proportional to x, t_x**
- Magnetization vector (rotating within XY plane):
 $\mathbf{N}_{xy}(x, y) \exp(i w_0 t) \exp(i G_y y t_y) \exp(i G_x x t_x)$

MRI

- **Step 5: Measure magnetization vector for entire volume**
 - Superposition of magnetization from each **$dxdy$** element within selected slice
 - **Integrate** over XY plane: $N_{xy}(x,y) \exp(i \omega_0 t) \exp(i G_y y t_y) \exp(i G_x x t_x) dxdy$
 - At discrete **timepoints** t_y (fixed) and t_x (varying), measurements equal **Fourier transform** of magnetization $N_{xy}(x,y) \exp(i \omega_0 t)$ evaluated at discrete “frequencies” ($G_y t_y, G_x t_x$), i.e., measurements at frequencies along a line/row in 2D “frequency” domain
- Repeat entire process at different discrete values of t_y to measure along a different line/row
- Repeat entire process until we fill required 2D frequency domain
- Acquired data of slice is Fourier transform of magnetization
- Reconstruct: Data $\rightarrow$ inverse Fourier transform $\rightarrow$ magnitude $\rightarrow N_{xy}(x,y)$

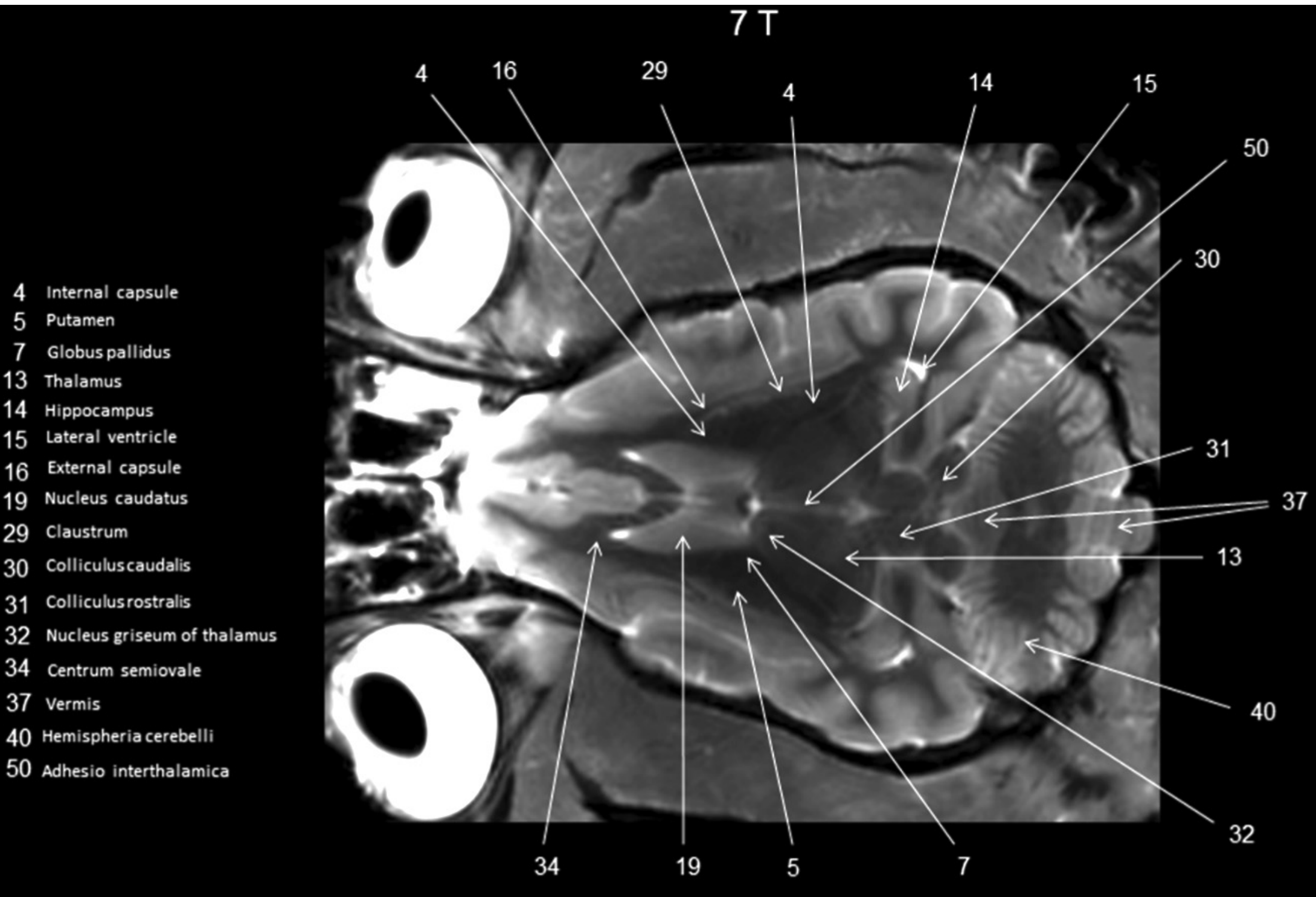
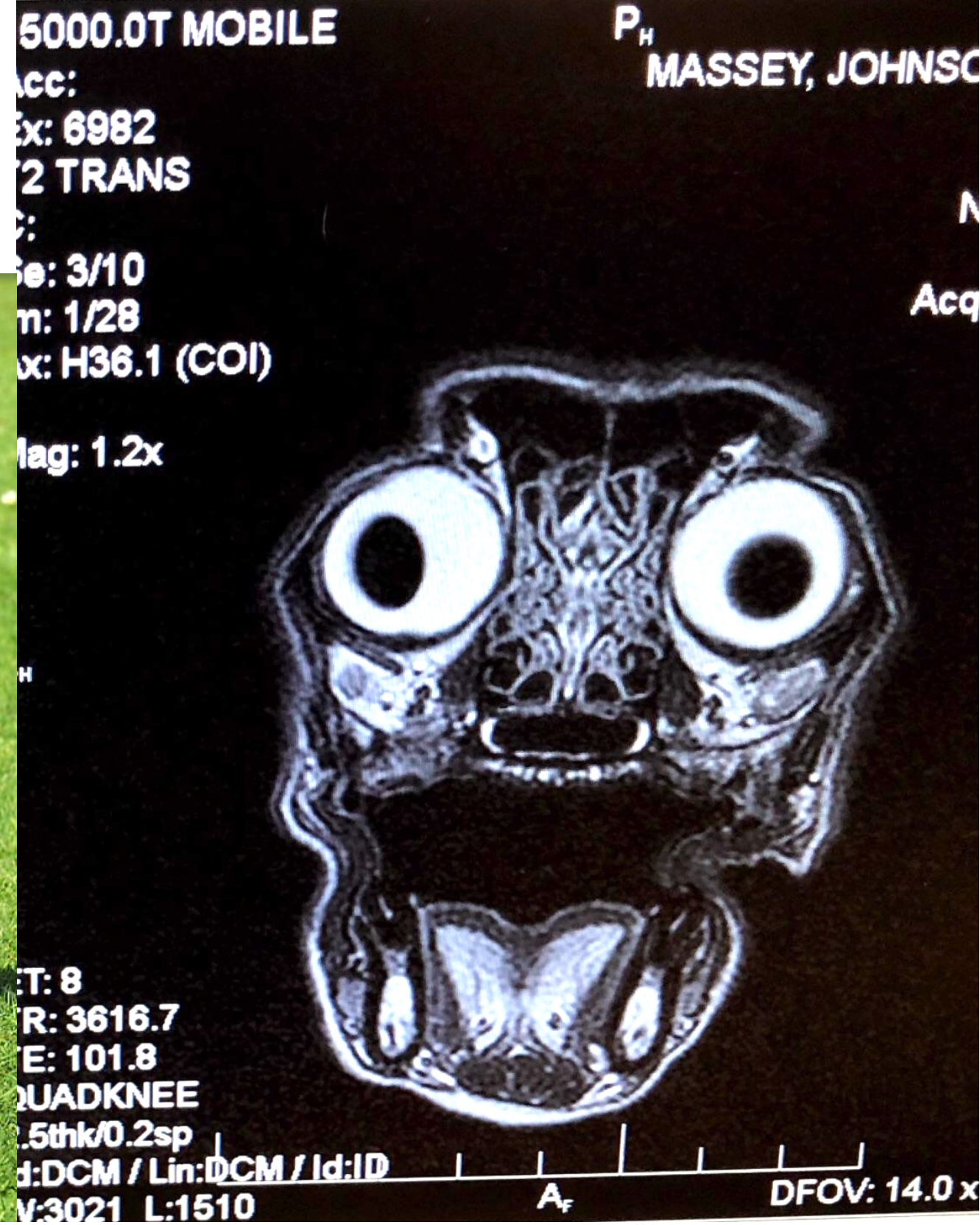


FIGURE 6 Dorsal T2-weighted turbo spin-echo image at 7T showing detailed anatomy of the canine brain

MRI

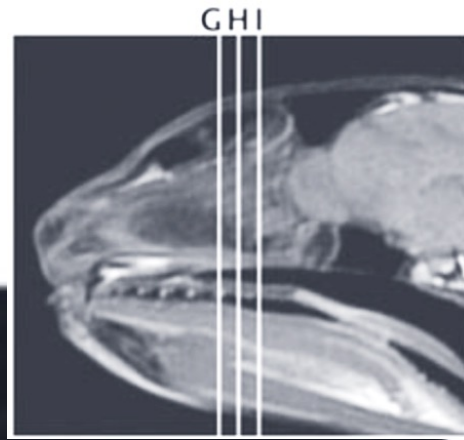
- MRI of a dog (dachshund) head



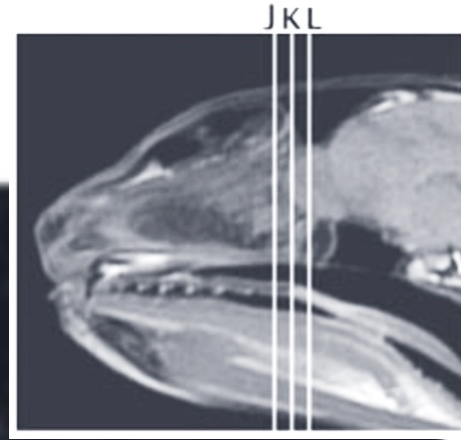
MRI

- MRI of a cat head

(G)



(J)



Magnetic Resonance Imaging of Normal Nasal Cavity and Paranasal Sinuses in Cats

F. Conchou¹, J. Sautet², F. Raharison³ and G. Mogicato^{2,4*}

MRI

- [Pareidolia](#)

Pareidolia in MRI: finding hidden patterns in medical imaging

Published 19 March 2025 Updated 1 April 2026 In MRI, Small Animal

Fig 1: Edvard Munch's "The Scream" shows here as a face in this transverse view of the nasal sinus

